

GENERAL POLYNOMIAL

An expression written as, $a_n x^n + a_{n-1} x^{n-1} + \dots + a_1 x + a_0$, where $a_0, a_1, a_2, \dots, a_n \in \mathbb{R}$ is called a **polynomial** of degree n where ($a_n \neq 0, n \in \mathbb{W}$).

If $a_0, a_1, a_2, \dots, a_n \in \mathbb{C}$, then it is called **complex coefficient polynomial**.

QUADRATIC POLYNOMIAL

$ax^2 + bx + c$, is a polynomial of degree two in one variable, where $a \neq 0$ & $a, b, c \in \mathbb{R}$. ($a \rightarrow$ leading coefficient, $c \rightarrow$ absolute term / constant term)

If $a = 0$ then above expression reduces to, $bx + c$, which known as **linear polynomial** ($b \neq 0$). If $c = 0$ in linear polynomial it reduces to, bx which is **odd linear polynomial**.

QUADRATIC EQUATIONS WITH REAL COEFFICIENTS

The expression, $ax^2 + bx + c$, where $a, b, c \in \mathbb{R}$ and $a \neq 0$ is called as a quadratic expression in x . The quadratic expression when equated to zero, $ax^2 + bx + c = 0$, is called as a quadratic equation in x . Where the numbers a, b, c are called the coefficients of the equation.

1. The values of x which satisfy the quadratic equation is called roots (also called solutions or zeros) of the quadratic equation.

2. This equation has two roots which are given by $x = \frac{-b \pm \sqrt{D}}{2a}$,

where D (or Δ) = $b^2 - 4ac$ is called as discriminant of the quadratic equation.



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Example 1: The roots of $3x^2 - 5x - 12 = 0$ are

Sol. $x = \frac{5 \pm \sqrt{25 - 4(3)(-12)}}{2 \times 3}$ i.e. $x = -\frac{4}{3}, 3$
or (using factorization method)

$$3x^2 - 9x + 4x - 12 = 0$$

$$3x(x - 3) + 4(x - 3) = 0$$

$$(3x + 4)(x - 3) = 0, x = -4/3, 3$$

Example 2: The quadratic equation whose roots are $1 + \sqrt{2}, 1 - \sqrt{2}$ is

Sol. $x^2 - (1 + \sqrt{2} + 1 - \sqrt{2})x + (1 + \sqrt{2})(1 - \sqrt{2}) = 0$

i.e. $x^2 - 2x + (-1) = 0$

$$x^2 - 2x - 1 = 0 \text{ (or) } x = 1 + \sqrt{2}$$

$$x = 1 - \sqrt{2} \text{ (or) } x = 1 - \sqrt{2}$$

$$x^2 - 2x - 1 = 0$$



Concept Application

1. If $3^{1+x} + 3^{1-x} = 10$ then the values of x are

- (a) 1, -1 (b) 1, 0
(c) 1, 2 (d) -1, -2

2. If a, b, c are positive real numbers then the number of real roots of the equation $ax^2 + b|x| + c = 0$ is

- (a) 2 (b) 4
(c) 0 (d) -1

RELATION BETWEEN ROOTS AND CO-EFFICIENTS

(i) The solutions of quadratic equation, $ax^2 + bx + c = 0$,

($a \neq 0$) are given by $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$

(ii) If α, β are the roots of quadratic equation,

$$ax^2 + bx + c = 0$$

then equation (i) can be written as

$$a(x - \alpha)(x - \beta) = 0$$

...(i)

$$\text{or } ax^2 - a(\alpha + \beta)x + a\alpha\beta = 0$$

...(ii)

equations (i) and (ii) are identical,

$\therefore$ by comparing the coefficients sum of the roots,

$$\alpha + \beta = -\frac{b}{a} = -\frac{\text{constant term}}{\text{coefficient of } x^2}$$

and product of the roots,

$$\alpha\beta = \frac{c}{a} = \frac{\text{coefficient of } x}{\text{coefficient of } x^2}$$

(iii) Dividing the equation (i) by a ,

$$x^2 + \frac{b}{a}x + \frac{c}{a} = 0$$

$$\Rightarrow x^2 - \left(\frac{-b}{a}\right)x + \frac{c}{a} = 0$$

$$\Rightarrow x^2 - (\alpha + \beta)x + \alpha\beta = 0$$

$$\Rightarrow x^2 - (\text{sum of the roots})x + (\text{product of the roots}) = 0$$

Hence we conclude that the quadratic equation whose roots are α & β is $x^2 - (\alpha + \beta)x + \alpha\beta = 0$



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Example 3: If α and β are the roots of $ax^2 + bx + c = 0$ and γ, δ are those of $lx^2 + mx + n = 0$, find the equation whose roots are $\alpha\gamma + \beta\delta$ and $\alpha\delta + \beta\gamma$.

Sol. $ax^2 + bx + c = 0$ has roots a and b

$$\therefore \alpha + \beta = -\frac{b}{a}; \quad \alpha\beta = \frac{c}{a}$$

$lx^2 + mx + n = 0$ has roots γ and δ

$$\therefore \gamma + \delta = -\frac{m}{l}; \quad \gamma\delta = \frac{n}{l}$$

The required quadratic equation is

$$x^2 - (\alpha\gamma + \beta\delta + \alpha\delta + \beta\gamma)x + (\alpha\gamma + \beta\delta)(\alpha\delta + \beta\gamma) = 0$$

$$\text{where } \alpha\gamma + \beta\delta + \alpha\delta + \beta\gamma = (\alpha + \beta)(\gamma + \delta) = \frac{b}{a} \cdot \frac{m}{l}$$

$$\begin{aligned} (\alpha\gamma + \beta\delta)(\alpha\delta + \beta\gamma) &= \frac{n}{l} \left(\frac{b^2 - 2ac}{a^2} \right) + \frac{c}{a} \left(\frac{m^2 - 2nl}{l^2} \right) \\ &= \frac{ln(b^2 - 2ac) + ac(m^2 - 2nl)}{a^2l^2} \end{aligned}$$

$$\therefore a^2l^2 \cdot x^2 - ab/mx + ln(b^2 - 2ac) + ac(m^2 - 2nl) = 0$$

Example 4: The quadratic equation whose roots are $\frac{3 \pm i\sqrt{5}}{2}$ is

$$\text{Sol. } x^2 - \left(\frac{3+i\sqrt{5}}{2} + \frac{3-i\sqrt{5}}{2} \right)x + \left(\frac{3+i\sqrt{5}}{2} \cdot \frac{3-i\sqrt{5}}{2} \right) = 0$$

$$x^2 - 3x + \frac{7}{2} = 0 \Rightarrow 2x^2 - 6x + 7 = 0$$

$$\text{or } x = \frac{3 \pm i\sqrt{5}}{2} \Rightarrow (2x - 3)^2 = -5$$

$$\Rightarrow 2x^2 - 6x + 7 = 0$$

Example 5: If α, β are the roots of $ax^2 + bx + c = 0$ and $c \neq 0$ then the value of $\frac{1}{(\alpha\alpha + b)^2} + \frac{1}{(\alpha\beta + b)^2}$ in terms of a, b, c is

Sol. $\because \alpha$ is a root of $ax^2 + bx + c = 0$

$$\Rightarrow a\alpha^2 + b\alpha = -c$$

$$\Rightarrow a\alpha + b = \frac{-c}{\alpha}, \text{ similarly } a\beta + b = \frac{-c}{\beta}$$

$$\therefore \frac{1}{(\alpha\alpha + b)^2} + \frac{1}{(\alpha\beta + b)^2} = \frac{(\alpha + \beta)^2 - 2\alpha\beta}{c^2} = \frac{b^2 - 2ac}{a^2c^2}$$



Concept Application

3. If α, β are roots of the equation $2x^2 + 6x + b = 0 (b < 0)$

then $\frac{\alpha}{\beta} + \frac{\beta}{\alpha}$ is less than

- (a) 2 (b) -2
(c) 18 (d) 0

4. The quadratic equation whose roots are

$$\frac{\sqrt{2} + \sqrt{3}i}{\sqrt{2} - \sqrt{3}i} \text{ and } \frac{\sqrt{2} - \sqrt{3}i}{\sqrt{2} + \sqrt{3}i}$$

- (a) $5x^2 - 2x + 5 = 0$ (b) $5x^2 + 2x + 5 = 0$
(c) $5x^2 + 2x - 5 = 0$ (d) $5x^2 - 2x - 5 = 0$

SYMMETRIC FUNCTION OF THE ROOTS

A function of α and β is said to be a symmetric function if it remains unchanged when α and β are interchanged.

For example, $\alpha^2 + \beta^2 + 2\alpha\beta$ is a symmetric function of α and β whereas $\alpha^2 - \beta^2 + 3\alpha\beta$ is not a symmetric function of α and β .

In order to find the value of a symmetric function of α and β , express the given function in terms of $\alpha + \beta$ and $\alpha\beta$. The following results may be useful.

$$(i) \alpha^2 + \beta^2 = (\alpha + \beta)^2 - 2\alpha\beta$$

$$(ii) \alpha^3 + \beta^3 = (\alpha + \beta)^3 - 3\alpha\beta(\alpha + \beta)$$

$$(iii) \alpha^4 + \beta^4 = (\alpha^3 + \beta^3)(\alpha + \beta) - \alpha\beta(\alpha^2 + \beta^2)$$

$$(iv) \alpha^5 + \beta^5 = (\alpha^3 + \beta^3)(\alpha^2 + \beta^2) - \alpha^2\beta^2(\alpha + \beta)$$

$$(v) |\alpha - \beta| = \sqrt{(\alpha + \beta)^2 - 4\alpha\beta}$$

$$(vi) \alpha^2 - \beta^2 = (\alpha + \beta)(\alpha - \beta)$$

$$(vii) \alpha^3 - \beta^3 = (\alpha - \beta)[(\alpha + \beta)^2 - \alpha\beta]$$

$$(viii) \alpha^4 - \beta^4 = (\alpha + \beta)(\alpha - \beta)(\alpha^2 + \beta^2)$$



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Example 6: If α, β are the root of $ax^2 + bx + c = 0$, then the equation whose roots are $2 + \alpha$ and $2 + \beta$ is

Sol. $f(x - 2) = 0 \Rightarrow a(x - 2)^2 + b(x - 2) + c = 0$
 $\Rightarrow ax^2 + x(b - 4a) + 4a - 2b + c = 0$

Example 7: If each root of the equation $x^2 + 11x + 13 = 0$ is diminished by 4, then the transferred equation is

Sol. $f(x + 4) = 0 \Rightarrow (x - 4)^2 + 11(x + 4) + 13 = 0$
 $\Rightarrow x^2 - 19x + 73 = 0$

Example 8: If α, β are the roots of $4x^2 + 7x + 2 = 0$, then the equation whose roots are α^2, β^2 is

Sol. $f(x) = 4x^2 + 7x + 2 = 0$
 The required equation is $f(\sqrt{x}) = 0$.

$$\Rightarrow 4(\sqrt{x})^2 + 7\sqrt{x} + 2 = 0 \Rightarrow 4x + 2 = -7\sqrt{x}$$

$$\Rightarrow (4x + 2)^2 = (-7\sqrt{x})^2 \Rightarrow 16x^2 - 33x + 4 = 0$$



Concept Application

5. If α, β are the roots of the equation $x^2 + 3x + 2$. Then $\alpha^5 + \beta^5 =$

(a) 33

(b) -32

(c) 35

(d) -33

NATURE OF ROOTS

❖ Consider the quadratic equation $ax^2 + bx + c = 0$ where $a, b, c \in \mathbb{R}$ & $a \neq 0$ then;

(i) $D > 0 \Leftrightarrow$ roots are real & distinct (unequal)

(ii) $D = 0 \Leftrightarrow$ roots are real & coincident (equal)

(iii) $D < 0 \Leftrightarrow$ roots are imaginary

(iv) If $p + iq$ is one root of a quadratic equation, then the other must be its conjugate $p - iq$ & vice versa. ($p, q \in \mathbb{R}$ & $i = \sqrt{-1}$)

❖ Consider the quadratic equation $ax^2 + bx + c = 0$ where $a, b, c \in \mathbb{Q}$ & $a \neq 0$ then ;

(i) If $D > 0$ & is a perfect square, then roots are rational & unequal.

(ii) If $\alpha = p + \sqrt{q}$ is one root in this case, (where p is rational & $\sqrt{q}$ is a surd) then the other root must be the conjugate of it i.e. $\beta = p - \sqrt{q}$ & vice versa.

$$\text{If } \Delta = b^2 - 4ac > 0.$$

Now consider the following cases.

Case		Nature of roots
Case-I	$a > 0, b > 0, c > 0$ $\Rightarrow \alpha + \beta < 0, \alpha\beta > 0$	Both roots are negative.
Case-II	$a > 0, b > 0, c < 0$ $\Rightarrow \alpha + \beta < 0, \alpha\beta < 0$	Both roots are opposite in sign; Magnitude of negative root is more than the magnitude of positive root.
Case-III	$a > 0, b < 0, c > 0$ $\Rightarrow \alpha + \beta > 0, \alpha\beta > 0$	Both roots are positive.
Case-IV	$a > 0, b < 0, c < 0$ $\Rightarrow \alpha + \beta > 0, \alpha\beta < 0$	Roots are opposite in sign. Magnitude of positive root is more than magnitude of negative root.

Remember

- (i) Roots are rational $\Leftrightarrow D$ is a perfect square.
- (ii) Roots are irrational $\Leftrightarrow D$ is positive but not a perfect square.
- (iii) If $a + b + c = 0$, then 1 is a root of the equation $ax^2 + bx + c = 0$.
- (iv) If a and c are of opposite sign, the roots must be of opposite sign.
- (v) If the roots are equal in magnitude but opposite in sign, then $b = 0, ac < 0$.
- (vi) If the roots are reciprocal of each other, then $c = a$.
- (vii) The quadratic equation whose roots are reciprocals of the roots of $ax^2 + bx + c = 0$ is $cx^2 + bx + a = 0$ (i.e., the coefficients are written in reverse order).
- (viii) If $a = 1, b, c \in \mathbb{Z}$ and the roots are rational numbers, then these roots must be integers.
- (ix) If $ax^2 + bx + c = 0$ is satisfied by more than two values, it is an identity and $a = b = c = 0$ and vice-versa.
- (x) If $P(x) = a_1x^2 + b_1x + c_1$ and $Q(x) = a_2x^2 + b_2x + c_2$, then $P(x) \cdot Q(x) = 0$ have atleast two real roots if $D_1 + D_2 > 0$.

Condition for a Root of a Quadratic Equation

$ax^2 + bx + c = 0$ to be reciprocal of the root of $a'x^2 + b'x + c' = 0$ is

$$(cc' - aa')^2 = (ba' - b'c)(ab' - bc')$$





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Example 9: Determine the values of m for which the equation $5x^2 - 4x + 2 + m(4x^2 - 2x - 1) = 0$ will have

- (a) equal roots
(b) product of the roots as 2

Sol. $5x^2 - 4x + 2 + m(4x^2 - 2x - 1) = 0$
 $(5 + 4m)x^2 - 2(m + 2)x + (2 - m) = 0 \quad \dots(i)$

(a) For equal roots, the discriminant of (i) should be equal to zero.

$$\begin{aligned} \therefore 4(m + 2)^2 - 4(5 + 4m)(2 - m) &= 0 \\ \Rightarrow m^2 + 4m + 4 - (10 + 8m - 5m - 4m^2) &= 0 \\ \Rightarrow 5m^2 + m - 6 &= 0 \\ \Rightarrow (5m + 6)(m - 1) &= 0 \\ \Rightarrow m = -\frac{6}{5} \text{ or } m &= 1 \end{aligned}$$

(b) $\frac{2 - m}{5 + 4m} = 2$

$$\therefore 2 - m = 10 + 8m \Rightarrow m = -\frac{8}{9}$$

Example 10: The nature of the roots of $x^2 + x - 1 = 0$ is

Sol. $\Delta = (1)^2 - 4(1)(-1) = 5$ is not a perfect square.
 $\Rightarrow$ roots are irrational and more over conjugate

Example 11: The nature of the roots of $2x^2 + x + 6 = 0$ is

Sol. $\Delta = 1 - 4(1)(-1) = 5$ is not a perfect square.
 $\Rightarrow$ roots are irrational and more over conjugate complex number

Example 12: If the roots of the equation $x^2 - 15 - m(2x - 8) = 0$ are equal then the value of m is

Sol. $x^2 - 15 - m(2x - 8) = 0$
 $x^2 - 2x + 8m - 15 = 0$
 $\Delta = 0 \Rightarrow (-2m)^2 - 4 \times 1 \times (8m - 15) = 0$
 $m^2 - 8m + 15 = 0$
 $(m - 3)(m - 5) = 0, m = 3; m = 5$

Example 13: If a, b, c are real, the roots of $a^2x^2 + b^2x^2 - 2abx - 2bcx + b^2 + c^2 = 0$ are real and equal, then the progression in which a, b, c lies is

Sol. $a^2x^2 + b^2x^2 - 2abx - 2bcx + b^2 + c^2 = 0$
 $a^2x^2 - 2abx + b^2 + b^2x^2 - 2bcx + c^2 = 0$
 $(ax - b)^2 + (bx - c)^2 = 0$
 $ax - b = 0; bx - c = 0$
 $x = \frac{b}{a}; x = \frac{c}{b}$
 $\frac{b}{a} = \frac{c}{b} \Rightarrow b^2 = ac$
 $\therefore a, b, c$ are in G.P.

Example 14: If the roots $(a^2 - bc)x^2 + 2(b^2 - ca)x + (c^2 - ab) = 0$ are equal then the condition (s) is (are)

Sol. $(a^2 - bc)x^2 + 2(b^2 - ca)x + (c^2 - ab) = 0$
 roots are equal $\Rightarrow \Delta = 0$
 $\Rightarrow 4(b^2 - ca)^2 - 4(a^2 - bc)(c^2 - ab) = 0$
 $\Rightarrow b^4 - 2ab^2c + a^3b + bc^3 - ab^2c = 0$
 $\Rightarrow b = 0$ (or) $a^3 + b^3 + c^3 - 3abc = 0$

Example 15: A quadratic equation $p(x) = 0$ has $1 + \sqrt{5}$ and $1 - \sqrt{5}$ as roots and it satisfies $p(1) = 2$. Find the quadratic polynomial.

Sol. Sum of the roots = 2, product of the roots = -4
 $\therefore$ let $p(x) = a(x^2 - 2x - 4) \Rightarrow p(1) = 2$
 $\Rightarrow 2 = a(1^2 - 2 \cdot 1 - 4) \Rightarrow a = -2/5$
 $\therefore p(x) = -2/5(x^2 - 2x - 4)$

Example 16: The quadratic equation $x^2 + mx + n = 0$ has roots which are as twice as those of $x^2 + px + m = 0$ and m, n and $p \neq 0$. Find the value of $\frac{n}{p}$.

Sol. $x^2 + mx + n = 0$ $\begin{cases} 2\alpha \\ 2\beta \end{cases}$
 and $x^2 + px + m = 0$ $\begin{cases} \alpha \\ \beta \end{cases}$

$$\begin{aligned} 2(\alpha + \beta) &= -m & \dots(i) \\ 4\alpha\beta &= n & \dots(ii) \\ \text{and } \alpha + \beta &= -p & \dots(iii) \\ \alpha\beta &= m & \dots(iv) \end{aligned}$$

$$\begin{aligned} \therefore \text{from (i) and (iii)} \\ \Rightarrow 2p &= m \\ \text{and from (ii) and (iv)} \\ \Rightarrow 4m &= n \Rightarrow \frac{n}{p} = \frac{4m}{m/2} = 8 \end{aligned}$$

Example 17: For what values of m the equation $(1 + m)x^2 - 2(1 + 3m)x + (1 + 8m) = 0$ has equal roots.

Sol. Given equation is
 $(1 + m)x^2 - 2(1 + 3m)x + (1 + 8m) = 0$
 Roots of equation will be equal if, $D = 0$.
 $\Rightarrow 4(1 + 3m)^2 - 4(1 + m)(1 + 8m) = 0$
 $\Rightarrow 4(1 + 9m^2 + 6m - 1 - 9m - 8m^2) = 0$
 $\Rightarrow m^2 - 3m = 0$ or, $m(m - 3) = 0$
 $\therefore m = 0, 3$.

Example 18: Find all the integral values of a for which the quadratic equation $(x - a)(x - 10) + 1 = 0$ has integral roots.

Sol. Here the equation is $x^2 - (a + 10)x + 10a + 1 = 0$. For integral roots D should be a perfect square.
 $D = a^2 - 20a + 96$.

$$\Rightarrow D = (a - 10)^2 - 4$$

$$\Rightarrow 4 = (a - 10)^2 - D$$

If D is a perfect square means difference of two perfect square as 4 which is possible only when $(a - 10)^2 = 4$ and $D = 0$.

$$\Rightarrow (a - 10) = \pm 2; \Rightarrow a = 12, 8$$



Concept Application

6. If one roots of $x^2 - (3 + 2i)x + (1 + 3i) = 0$ is $1 + i$ then the other roots is

- (a) $1 - i$ (b) $2 + i$
(c) $3 + i$ (d) $1 + 3i$

7. If the roots of the quadratic equation $x^2 - 4x - \log_3 a = 0$ are real, then the least value of a is

- (a) 81 (b) $\frac{1}{81}$ (c) $\frac{1}{64}$ (d) 9

8. If the roots of the equation

$(a^2 + b^2)x^2 + 2(bc + ad)x + (c^2 + d^2) = 0$ are real then a^2, b^2, c^2 are in

- (a) A.P. (b) G.P.
(c) H.P. (d) A.G.P.

QUADRATIC EXPRESSION

$$\text{Let } y = ax^2 + bx + c = a \left(\left(x + \frac{b}{2a} \right)^2 - \frac{D}{4a^2} \right) \quad \dots(i)$$

1. Represents a parabola with vertex $\left(\frac{-b}{2a}, \frac{-D}{4a} \right)$ and axis of the

parabola is $x = \frac{-b}{2a}$

If $a > 0$, the parabola opens upward while if $a < 0$, the parabola opens downward. The parabola cuts the x -axis at points corresponding to roots of $ax^2 + bx + c = 0$. If this equation has

- (i) $D > 0$, the parabola cuts x -axis at two real and distinct points.

- (ii) $D = 0$, the parabola touches x -axis at $x = \frac{-b}{2a}$.

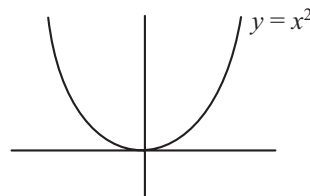
- (iii) $D < 0$, then;

if $a > 0$, parabola lies above x -axis.

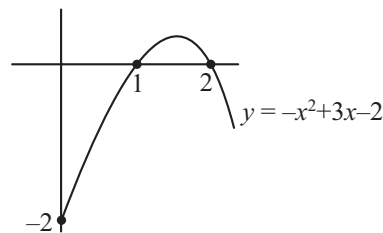
if $a < 0$, parabola lies below x -axis.

e.g.

1. $y = x^2$



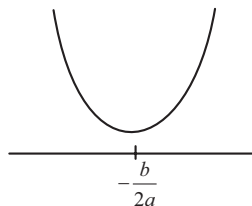
2. $y = -x^2 + 3x - 2$



GRAPHS OF QUADRATIC EXPRESSIONS

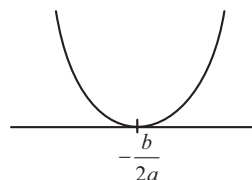
Let $f(x) = ax^2 + bx + c$ and $\alpha, \beta : \alpha < \beta$, be its roots

- (i) $a > 0$ and $D < 0 \Leftrightarrow f(x) > 0 \quad \forall x \in \mathbb{R}$



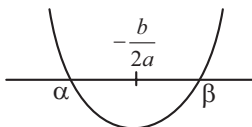
α and β are complex conjugates

- (ii) $a > 0$ and $D = 0 \Leftrightarrow f(x) \geq 0 \quad \forall x \in \mathbb{R}$



$$f(x) = 0 \text{ at } x = \frac{-b}{2a}$$

- (iii) $a > 0$ and $D > 0$; then,

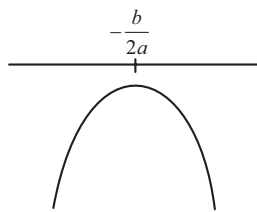


$$f(x) > 0 \quad \forall x \in (-\infty, \alpha) \cup (\beta, \infty)$$

$$f(x) < 0 \quad \forall x \in (\alpha, \beta)$$

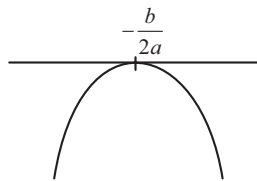
$$f(x)_{\min} = -\frac{D}{4a} \text{ at } x = -\frac{b}{2a}$$

- (iv) $a < 0$ and $D < 0 \Leftrightarrow f(x) < 0 \quad \forall x \in \mathbb{R}$



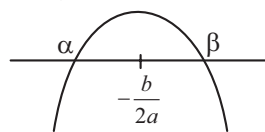
α and β are complex conjugates

(v) $a < 0$ and $D = 0 \Leftrightarrow f(x) \leq 0 \quad \forall \quad x \in \mathbb{R}$



$$f(x) = 0 \text{ at } x = -\frac{b}{2a}$$

(vi) $a < 0$ and $D > 0$; then



$$f(x) < 0 \quad \forall \quad x \in (-\infty, \alpha) \cup (\beta, \infty)$$

$$f(x) > 0 \quad \forall \quad x \in (\alpha, \beta)$$

$$f(x)_{\max} = -\frac{D}{4a} \text{ at } x = -\frac{b}{2a}$$

EQUATION V/S IDENTITY

A quadratic equation is satisfied by exactly two values of 'x' which may be real or imaginary. The equation, $ax^2 + bx + c = 0$ is: A **quadratic equation** if $a \neq 0$ and An **identity** if $a = b = c = 0$

Infinite Roots
If an equation is satisfied by three distinct values of 'x', then it is an identity.

$(x+1)^2 = x^2 + 2x + 1$ is an identity in x.

Here highest power of x in the given relation is 2 and this relation is satisfied by three different values $x = 0$, $x = 1$ and $x = -1$ and hence it is an identity because a polynomial equation of n^{th} degree cannot have more than n distinct roots.

THEORY OF EQUATIONS

If $\alpha_1, \alpha_2, \alpha_3, \dots, \alpha_n$ are the roots of the equation; $f(x) = a_0x^n + a_1x^{n-1} + a_2x^{n-2} + \dots + a_{n-1}x + a_n = 0$ where $a_0, a_1, \dots, a_n$ are all real & $a_0 \neq 0$ then,

$$\sum \alpha_1 = -\frac{a_1}{a_0}, \sum \alpha_1 \alpha_2 = +\frac{a_2}{a_0}, \sum \alpha_1 \alpha_2 \alpha_3 = -\frac{a_3}{a_0}, \dots, \alpha_1 \alpha_2 \alpha_3$$

$$\dots \alpha_n = (-1)^n \frac{a_n}{a_0}$$

Note:

- (i) If α is a root of the equation $f(x) = 0$, then the polynomial $f(x)$ is exactly divisible by $(x - \alpha)$ or $(x - \alpha)$ is a factor of $f(x)$ and conversely.

- (ii) Every equation of n^{th} degree ($n \geq 1$) has exactly n roots & if the equation has more than n roots, it is an identity.
- (iii) If the coefficients of the equation $f(x) = 0$ are all real and $\alpha + i\beta$ is its root, then $\alpha - i\beta$ is also a root. i.e. **imaginary roots occur in conjugate pairs**.
- (iv) If the coefficients in the equation are all rational & $\alpha + \sqrt{\beta}$ is one of its roots, then $\alpha - \sqrt{\beta}$ is also a root where $\alpha, \beta \in \mathbb{Q}$ & β is not a perfect square.
- (v) If there be any two real numbers 'a' & 'b' such that $f(a)$ & $f(b)$ are of opposite signs, then $f(x) = 0$ must have at least one real root between 'a' and 'b'.
- (vi) Every equation $f(x) = 0$ of degree odd has at least one real root of a sign opposite to that of its last term.



Train Your Brain

Example 19: If $a + b + c = 0$; $an^2 + bn + c = 0$ and $a + bn + cn^2 = 0$ where $n \neq 0, 1$, then prove that $a = b = c = 0$.

Sol. Note that $ax^2 + bx + c = 0$ is satisfied by $x = 1$;

$$x = n \text{ \& } x = \frac{1}{n} \text{ where } n \neq \frac{1}{n}$$

$\Rightarrow$ Q.E. has 3 distinct real roots which implies that it must be an identity.

Example 20: If $x = 1$ and $x = 2$ are solutions of the equation $x^3 + ax^2 + bx + c = 0$ and $a + b = 1$, then find the value of b.

Sol. $a + b + c = -1$

$$\Rightarrow c = -2 \text{ \& } 8 + 4a + 2b + c = 0$$

$$\Rightarrow 4a + 2b = -6$$

$$\Rightarrow 2a + b = -3$$

$$\Rightarrow a = -4, b = 5$$

Example 21: A polynomial in x of degree greater than 3 leaves the remainder 2, 1 and -1 when divided by $(x - 1)$; $(x + 2)$ & $(x + 1)$ respectively. Find the remainder, if the polynomial is divided by, $(x^2 - 1)(x + 2)$.

Sol. $f(x) = Q_1(x - 1) + 2 = Q_2(x + 2) + 1 = Q_3(x + 1) - 1$

$$\Rightarrow f(1) = 2; f(-2) = 1; f(-1) = -1$$

$$\text{Let } f(x) = Q_r(x^2 - 1)(x + 2) + ax^2 + bx + c$$

$$\text{Hence } a + b + c = 2; 4a - 2b + c = 1$$

$$\text{and } a - b + c = -1$$

Example 22: The roots of the equation $a(b - c)x^2 + b(c - a)x + c(a - b) = 0$ ($a \neq b \neq c \neq 0$) are

Sol. $a(b - c) + b(c - a) + c(a - b) = 0$

$$\therefore \text{ the roots are } 1, \frac{c(a - b)}{a(b - c)}$$

Example 23: The roots of $ax^2 + 3bx + c = 0$ if $3b = a + c$ are

Sol. $ax^2 + 3bx + c = 0$ and $3b = a + c$

$$ax^2 + (a + c)x + c = 0; ax^2 + ax + cx + c = 0$$

$$ax(x + 1) + c(x + 1) = 0; (x + 1)(ax + c) = 0$$

$$x = -1; x = -\frac{c}{a}$$

$$\therefore \text{ roots are } -1, -\frac{c}{a}$$



Concept Application

9. If ω is an imaginary cube root of unity. Then the equation whose roots are $2\omega + 3\omega^2$, $2\omega^2 + 3\omega$ is

- (a) $x^2 + 5x + 7 = 0$ (b) $x^2 + 5x - 7 = 0$
(c) $x^2 - 5x + 7 = 0$ (d) $x^2 - 5x - 7 = 0$

COMMON ROOTS

One Root Common

If $\alpha \neq 0$ is a common root of the equation

$$a_1x^2 + b_1x + c_1 = 0 \quad \dots(i)$$

$$\text{and } a_2x^2 + b_2x + c_2 = 0 \quad \dots(ii)$$

Then we have

$$a_1\alpha^2 + b_1\alpha + c_1 = 0 \text{ and } a_2\alpha^2 + b_2\alpha + c_2 = 0$$

$$\text{These give } \frac{\alpha^2}{b_1c_2 - b_2c_1} = \frac{\alpha}{c_1a_2 - c_2a_1}$$

$$= \frac{1}{a_1b_2 - a_2b_1} (a_1b_2 - a_2b_1 \neq 0).$$

Thus, the required condition for one common root is $(a_1b_2 - a_2b_1)(b_1c_2 - b_2c_1) = (c_1a_2 - c_2a_1)^2$ and the value of the common root is

$$\alpha = \frac{c_1a_2 - c_2a_1}{a_1b_2 - a_2b_1} \text{ or } \frac{b_1c_2 - b_2c_1}{c_1a_2 - c_2a_1}.$$

Both Roots Common

If the equations (i) and (ii) have both roots common, then these equations will be identical. Thus the required condition for both roots common is

$$\frac{a_1}{a_2} = \frac{b_1}{b_2} = \frac{c_1}{c_2} \text{ (If no root is equal to zero)}$$

Remember

- (i) To find the common root of two equations, make the coefficient of second degree terms in two equations equal and subtract. The value of x so obtained is the required common root.
- (ii) If two quadratic equations with real and rational coefficients have a common imaginary or irrational root, then both roots will be common and the two equations will be identical. The required condition is

$$\frac{a_1}{a_2} = \frac{b_1}{b_2} = \frac{c_1}{c_2}.$$

- (iii) If α is a repeated root of the quadratic equation

$$f(x) = ax^2 + bx + c = 0,$$

Then α is also a root of the equation $f'(x) = 0$.

Note: If $f(x) = 0$ & $g(x) = 0$ are two polynomial equation having some common root(s) then those common root(s) is/are also the root(s) of $h(x) \equiv a f(x) + b g(x) = 0$.



Train Your Brain

Example 24: Find m and n in order that the equations $mx^2 + 5x + 2 = 0$ and $3x^2 + 10x + n = 0$ may have both the roots common.

Sol. The equations are $mx^2 + 5x + 2 = 0$ and $3x^2 + 10x + n = 0$. Since they have both the roots common,

$$\frac{m}{3} = \frac{5}{10} = \frac{2}{n} \therefore \frac{a_1}{a_2} = \frac{b_1}{b_2} = \frac{c_1}{c_2}$$

$$\text{From the first-relation, } m = \frac{15}{10} = \frac{3}{2}.$$

$$\text{From the last-relation, } n = 4.$$

Example 25: If the quadratic equations, $x^2 + bx + c = 0$ and $bx^2 + cx + 1 = 0$ have a common root then prove that either $b + c + 1 = 0$ or $b^2 + c^2 + 1 = bc + b + c$.

$$\text{Sol. } \alpha^2 + b\alpha + c = 0 \quad \dots(i)$$

$$b\alpha^2 + c\alpha + 1 = 0 \quad \dots(ii)$$

$$\therefore \frac{\alpha^2}{b - c^2} = \frac{\alpha}{bc - 1} = \frac{1}{c - b^2}$$

$$\Rightarrow \alpha^2 = \frac{b - c^2}{c - b^2} \text{ or } \alpha = \frac{bc - 1}{c - b^2}$$

$$\Rightarrow (bc - 1)^2 = (b - c^2)(c - b^2)$$

$$\text{or } b^3 + c^3 + 1 - 3bc = 0$$

$$\Rightarrow (b + c + 1)(b^2 + c^2 + 1 - bc - c - b) = 0$$

$$\Rightarrow b + c + 1 = 0 \text{ or } b^2 + c^2 + 1 = bc + b + c$$

Example 26: If $x^2 + 4ax + 3 = 0$ and $2x^2 + 3ax - 9 = 0$ have a common root then the value of a is

$$\text{Sol. } (c_1a_2 - c_2a_1)^2 = (a_1b_2 - a_2b_1)(b_1c_2 - b_2c_1)$$

$$(6 + 9)^2 = (3a - 8a)(-36a) \Rightarrow a = \pm 1$$

Example 27: If $x^2 + bx + c = 0$, $x^2 + cx + b = 0$ ($b \neq c$), have a common root then the condition is

Sol. Let α be the common root

$$\Rightarrow \alpha^2 + b\alpha + c = 0 \quad \dots(i)$$

$$\alpha^2 + c\alpha + b = 0 \quad \dots(ii)$$

$$(i) - (ii) \Rightarrow \alpha = 0$$

$$\text{Substitute in (i)}$$

$$\Rightarrow b + c + 1 = 0$$

Example 28: If the equation $k(6x^2 + 3) + rx + (2x^2 - 1) = 0$ and $6k(2x^2 + 1) + px + (4x^2 - 2) = 0$ have both roots common, then the value of p/r is

$$\text{Sol. } (6k + 2)x^2 + rs + 3k - 1 = 0 \quad \dots(i)$$

$$(12k + 4)x^2 + px + 6k - 2 = 0 \quad \dots(ii)$$

Since the roots of (i) and (ii) are equal,

$$\frac{6k + 2}{12k + 4} = \frac{r}{p} = \frac{3k - 1}{6k - 2} \therefore \frac{r}{p} = \frac{1}{2}$$





Concept Application

10. If the equation $x^2 + ax + b = 0$ and $x^2 + bx + a = 0$ ($a \neq b$) have a common root then $a + b =$

- (a) -1 (b) 0
(c) 1 (d) 2

RANGE OF QUADRATIC EXPRESSION

$$f(x) = ax^2 + bx + c$$

(i) Range when $x \in \mathbf{R}$: If $a > 0 \Rightarrow f(x) \in \left[-\frac{D}{4a}, \infty\right)$

$$a < 0 \Rightarrow f(x) \in \left(-\infty, \frac{D}{4a}\right]$$

Maximum & Minimum Value of $y = ax^2 + bx + c$ occurs at $x = -(b/2a)$ according as $a < 0$ or $a > 0$ respectively

(ii) Range in restricted domain : Given $x \in [x_1, x_2]$

(a) If $-\frac{b}{2a} \notin [x_1, x_2]$ then,

$$f(x) \in [\min\{f(x_1), f(x_2)\}, \max\{f(x_1), f(x_2)\}]$$

(b) If $-\frac{b}{2a} \in [x_1, x_2]$ then,

$$f(x) \in \left[\min\left\{f(x_1), f(x_2), -\frac{D}{4a}\right\}, \max\left\{f(x_1), f(x_2), -\frac{D}{4a}\right\}\right]$$



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Example 29: Find the minimum value of $f(x) = x^2 - 5x + 6$.

Sol. Minimum value of $f(x) = -\frac{D}{4a} = -\left(\frac{25-24}{4}\right)$
 $= -\frac{1}{4}$ (at $x = -\frac{b}{2a} = \frac{5}{2}$). Hence range is $\left[-\frac{1}{4}, \infty\right)$.

Example 30: Find the range of $y = \frac{x+2}{2x^2+3x+6}$, if x is real.

Sol. $y = \frac{x+2}{2x^2+3x+6}$
 $\Rightarrow 2yx^2 + 3yx + 6y = x + 2$
 $\Rightarrow 2yx^2 + (3y-1)x + 6y-2 = 0 \quad \dots(i)$

Case-I: If $y \neq 0$, then equation (i) is quadratic in x

$\therefore x$ is real

$\therefore D \geq 0$

$$\Rightarrow (3y-1)^2 - 8y(6y-2) \geq 0$$

$$\Rightarrow (3y-1)(13y+1) \leq 0$$

$$y \in \left[-\frac{1}{13}, \frac{1}{3}\right] - \{0\}$$

Case-II: if $y = 0$, then equation becomes

$$x = -2$$

Which is possible as x is real

$$\therefore \text{Range } y \in \left[-\frac{1}{13}, \frac{1}{3}\right]$$

Example 31: The maximum or minimum values of $2x + 5 - 3x^2$ on R is

Sol. Given expression $2x + 5 - 3x^2$,

$$a = -3 < 0, b = 2, c = 5$$

$\Rightarrow$ given expression has maximum value

$$\therefore \text{maximum value} = \frac{4ac - b^2}{4a} = \frac{4(-3)(5) - 4}{4(-3)} = \frac{16}{-3}$$



Concept Application

11. The maximum value of $\frac{1}{4x^2 + 2x + 1}$ is

- (a) $\frac{4}{3}$ (b) $\frac{2}{3}$ (c) 1 (d) $\frac{3}{4}$

12. If $x \in [2, 4]$ then for the expression $x^2 - 6x + 5$

- (a) The least value = 4
 (b) The greatest value = 4
 (c) The least value = 3
 (d) The greatest value = -5

13. Find the range of the following functions:

- (i) $f(x) = x^2 + x + 1$ (ii) $f(x) = 2x^2 + 3x + 5$
 (iii) $f(x) = x^2 + 2x + 7$ (iv) $f(x) = (x-1)(x-2)$
 (v) $f(x) = \frac{(x-1)}{x^2 + x + 1}$ (vi) $f(x) = \frac{2x^2 + 3x + 1}{x^2 + x + 1}$
 (vii) $f(x) = \frac{(x-1)(x-2)}{(x-3)}$ (viii) $f(x) = \frac{(x-1)(x-2)}{(x-3)(x-4)}$
 (ix) $f(x) = \frac{(x-1)(x-3)}{(x-2)(x-4)}$ (x) $f(x) = \frac{x^2 - 5}{2x^2 + 3}$
 (xi) $f(x) = x^2 + x + 1; x \geq 0$ (xii) $f(x) = 2x^2 + x + 1; 1 \leq x \leq 5$
 (xiii) $f(x) = \sin 4x + \cos^4 x$ (xiv) $f(x) = \sin^6 x + \cos^6 x$

SOLUTION OF QUADRATIC INEQUALITIES

The values of 'x' satisfying the inequality, $ax^2 + bx + c > 0$ ($a \neq 0$) are:

- (i) If $D > 0$, i.e. the equation $ax^2 + bx + c = 0$ has two different roots $\alpha < \beta$.
 $a > 0 \Rightarrow x \in (\infty, \alpha) \cup (\beta, \infty)$
 $a < 0 \Rightarrow x \in (\alpha, \beta)$
- (ii) If $D = 0$, i.e. roots are equal, i.e. $\alpha = \beta$.
 $a > 0 \Rightarrow x \in (-\infty, \alpha) \cup (\alpha, \infty)$
 $a < 0 \Rightarrow x \in \phi$
- (iii) If $D < 0$, i.e. the equation $ax^2 + bx + c = 0$ has no real roots.
 $a > 0 \Rightarrow x \in \mathbb{R}$
 $a < 0 \Rightarrow x \in \phi$

- (iv) Inequalities of the form $\frac{P(x)}{Q(x)} \leq 0$ can be solved using the method of intervals.



Train Your Brain

Example 32: Find the solution set of k so that $y = kx$ is secant to the curve $y = x^2 + k$.

Sol. Put $y = kx$ in $y = x^2 + k$

$$\begin{aligned} \Rightarrow kx &= x^2 + k = 0 \\ \Rightarrow x^2 - kx + k &= 0 \text{ for line to be secant, } D > 0 \\ \Rightarrow k^2 - 4k > 0 \quad k(k-4) > 0 \\ \text{hence } k &> 4 \text{ or } k < 0 \\ \Rightarrow k &\in (-\infty, 0) \cup (4, \infty) \end{aligned}$$

Example 33: Solve inequality $\frac{x^2 + x + 1}{|x + 1|} > 0$.

Sol. $\frac{x^2 + x + 1}{|x + 1|} > 0 \quad \dots(i)$

$$\begin{aligned} \therefore |x + 1| > 0, \forall x \in \mathbb{R} - \{-1\} \\ \therefore (i) \text{ becomes } x^2 + x + 1 > 0 \quad x \neq -1 \\ \therefore x^2 + x + 1 > 0, \forall x \in \mathbb{R} \text{ (as its } a > 0 \text{ and } D < 0) \\ \therefore \text{Solution of (i) is } x \in \mathbb{R} - \{-1\} \end{aligned}$$

Example 34: Solve $\left| \frac{x^2 - 3x - 1}{x^2 + x + 1} \right| < 3$.

Sol. $\left| \frac{x^2 - 3x - 1}{x^2 + x + 1} \right| < 3$

$$\begin{aligned} \therefore (x^2 + x + 1) > 0, \forall x \in \mathbb{R} \\ \therefore |x^2 - 3x - 1| < 3(x^2 + x + 1) \\ \Rightarrow (x^2 - 3x - 1)^2 - \{3(x^2 + x + 1)\}^2 < 0 \\ \Rightarrow (4x^2 + 2)(-2x^2 - 6x - 4) < 0 \\ \Rightarrow (2x^2 + 1)(x + 2)(x + 1) > 0 \\ \Rightarrow x \in (-\infty, -2) \cup (-1, \infty) \end{aligned}$$



Concept Application

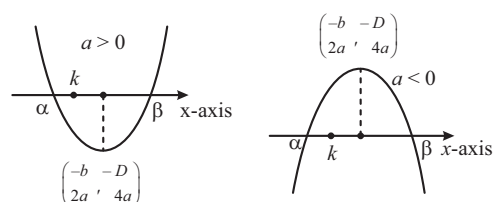
14. The set of values of x for which the inequality $[x]^2 - 5[x] + 6 \leq 0$ (where $[.]$ denote the greatest function) hold good is

- (a) $2 \leq [x] < 3$
- (b) $2 \leq x < 4$
- (c) $2 \leq x < 3$
- (d) $2 \leq x \leq 4$

LOCATION OF ROOTS

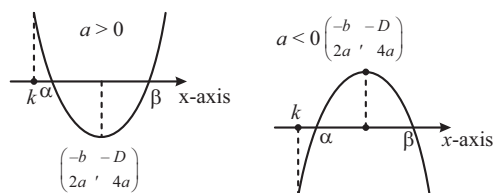
Let $f(x) = ax^2 + bx + c$, $a, b, c \in \mathbb{R}$, $a \neq 0$ and α, β be roots of $f(x) = 0$

1. A real number k lies between the roots of $f(x) = 0$



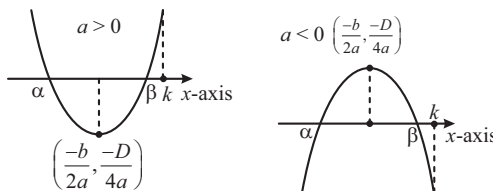
- (i) $D > 0$, (ii) $af(k) < 0$, where $\alpha < \beta$

2. If both roots of quadratic equation $f(x) = 0$ are greater than k, then



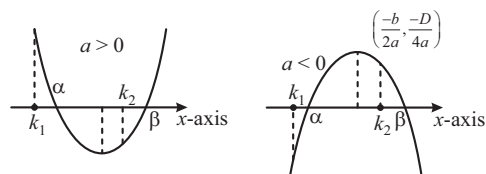
- (i) $D \geq 0$,
- (ii) $af(k) > 0$
- (iii) $k < \frac{-b}{2a}$, where $\alpha \leq \beta$

3. If both roots are less than real number k, then



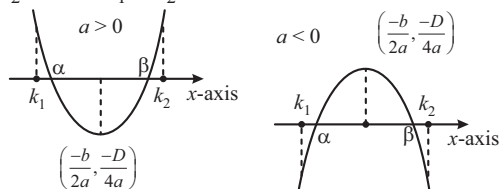
- (i) $D \geq 0$,
- (ii) $af(k) > 0$
- (iii) $k > \frac{-b}{2a}$, where $\alpha \leq \beta$

4. Exactly one root lies between real numbers k_1 and k_2 , where $k_1 < k_2$.



(i) $D > 0$, (ii) $f(k_1)f(k_2) < 0$, where $\alpha < \beta$

5. If both roots of $f(x) = 0$ are confined between real numbers k_1 and k_2 , where $k_1 < k_2$.



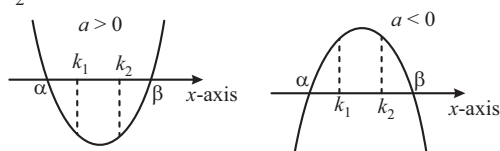
(i) $D \geq 0$,

(ii) $af(k_1) > 0$

(iii) $af(k_2) > 0$,

(iv) $k_1 < \frac{-b}{2a} < k_2$, where $\alpha \leq \beta$

6. If real numbers k_1 and k_2 lie between roots of $f(x) = 0$, where $k_1 < k_2$.



(i) $D > 0$,

(ii) $af(k_1) < 0$

(iii) $af(k_2) < 0$, where $\alpha < \beta$



Train Your Brain

Example 35: If α & β are the roots of $x^2 + px + q = 0$ and α^4, β^4 are the roots of $x^2 - rx + s = 0$, then the equation $x^2 - 4qx + 2q^2 - r = 0$ has always ($p, q, r, s \in \mathbb{R}$)

Sol. $\alpha + \beta = -p$; $\alpha\beta = q$; $\alpha^4 + \beta^4 = r$; $\alpha^4\beta^4 = s$

Now for equation $x^2 - 4qx + 2q^2 - r = 0$

$D = 16q^2 - 4(2q^2 - r)$

$= 8q^2 + 4r = 4(2q^2 + r) = 4[2\alpha^2\beta^2 + \alpha^4 + \beta^4]$

$= 4[(\alpha^2 + \beta^2)^2] = 4[(\alpha + \beta)^2 - 2\alpha\beta]^2 = 4(p^2 - 2q)^2 > 0$

Again consider the product of the roots $= 2q^2 - r$ which can be either positive or negative.

Example 36: Find the range of values of a for which the equation $x^2 - (a - 5)x + \left(a - \frac{15}{4}\right) = 0$ has at least one positive root.

Sol. $D \geq 0 \Rightarrow (a - 5)^2 - 4\left(a - \frac{15}{4}\right) \geq 0$

$\Rightarrow a^2 - 10a + 25 - 4a + 15 \geq 0 \Rightarrow a^2 - 14a + 40 \geq 0$

$\Rightarrow (a - 4)(a - 10) \geq 0$

$\Rightarrow a \in (-\infty, 4] \cup [10, \infty)$

Case I: When both roots are positive

$$D \geq 0, a - 5 > 0, a - \frac{15}{4} > 0$$

$$\Rightarrow D \geq 0, a > 5, a > \frac{15}{4} \Rightarrow a \in [10, \infty)$$

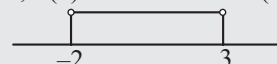
Case II: when exactly one root is positive

$$\Rightarrow a - \frac{15}{4} \leq 0 \Rightarrow a \leq \frac{15}{4}$$

then finally $a \in (-\infty, 15/4] \cup [10, \infty)$

Example 37: Find the value of k for which one root of the equation of $x^2 - (k + 1)x + k^2 + k - 8 = 0$ exceed 2 and other is smaller than 2.

Sol. Since $a > 0$, $f(0) < 0 \therefore f(2) < 0$



$$\therefore 4 - 2(k + 1) + k^2 + k - 8 < 0$$

$$\Rightarrow k^2 - k - 6 < 0$$

$$\Rightarrow (k + 2)(k - 3) < 0 \therefore k \in (-2, 3)$$

Example 38: Let $x^2 - (m - 3)x + m = 0$ ($m \in \mathbb{R}$) be a quadratic equation, then find the values of ' m ' for which

(a) both the roots are greater than 2.

(b) both roots are positive.

(c) one root is positive and other is negative.

(d) One root is greater than 2 and other smaller than 1

(e) Roots are equal in magnitude and opposite in sign.

(f) both roots lie in the interval (1, 2)

Sol. (a) $f(x)$



Condition I: $D \geq 0$

$$\Rightarrow (m - 3)^2 - 4m \geq 0$$

$$\Rightarrow m^2 - 10m + 9 \geq 0$$

$$\Rightarrow (m - 1)(m - 9) \geq 0$$

$$\Rightarrow m \in (-\infty, 1] \cup [9, \infty) \quad \dots(i)$$

Condition II: $f(2) > 0$

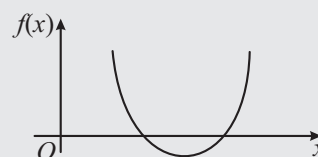
$$\Rightarrow 4 - (m - 3)2 + m > 0$$

$$\Rightarrow m < 10 \quad \dots(ii)$$

$$\text{Condition III: } -\frac{b}{2a} > 2 \Rightarrow m > 7 \quad \dots(iii)$$

Intersection of (i), (ii) and (iii) gives $m \in [9, 10)$

(b)



Condition I: $D \geq 0$

$$\Rightarrow m \in (-\infty, 1] \cup [9, \infty)$$

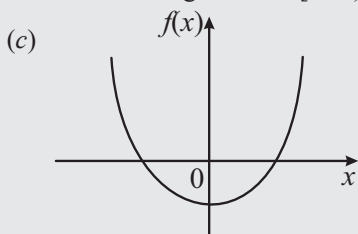
Condition II: $f(0) > 0$

$$\Rightarrow m > 0$$

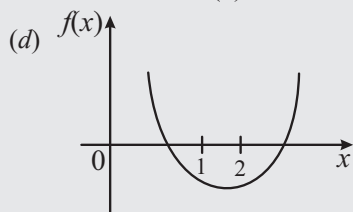
Condition III: $-\frac{b}{2a} > 0 \Rightarrow \frac{m-3}{2} > 0$

$$\Rightarrow m > 3$$

intersection gives $m \in [9, \infty)$ **Ans.**



Condition I: $f(0) < 0 \Rightarrow m < 0$ **Ans.**



Condition I: $f(1) < 0 \Rightarrow 4 < 0 \Rightarrow m \in \phi$

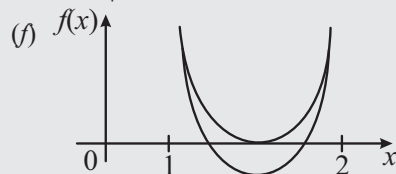
Condition II: $f(2) < 0 \Rightarrow m > 10$

Intersection gives $m \in \phi$

(e) sum of roots = 0 $\Rightarrow m = 3$

and $f(0) < 0 \Rightarrow m < 0$

$$\therefore m \in \phi.$$



Condition I: $D \geq 0 \Rightarrow m \in (-\infty, 1] \cup [9, \infty)$

Condition II: $f(1) > 0 \Rightarrow 1 - (m-3) + m > 0 \Rightarrow 4 > 0$ which is true $\forall m \in \mathbb{R}$

Condition III: $f(2) > 0$

$$\Rightarrow m < 10$$

Condition IV: $1 < -\frac{b}{2a} < 2$

$$\Rightarrow 1 < \frac{m-3}{2} < 2$$

$$\Rightarrow 5 < m < 7 \text{ intersection gives } m \in \phi$$

Example 39: If both roots of the quadratic equation $x^2 - 2kx + k^2 + K - 5 = 0$ are less than 5, then find the interval in which k lies

Sol. $\Delta \geq 0 \Rightarrow 4k^2 - 4(k^2 + k - 5) \geq 0 \Rightarrow k \leq 5$... (i)

$$af(k) \geq 0 \Rightarrow 1. f(5) > 0$$

$$\Rightarrow k^2 - 9k + 20 > 0$$

$$\Rightarrow k < 4 \text{ or } k > 5$$
 ... (ii)

$$k > \frac{-b}{2a} \Rightarrow 5 > \frac{-(-2k)}{2} \Rightarrow k < 5$$
(iii)

from (i), (ii) and (iii)

$$k \in (-\infty, 4).$$

Example 40: The interval in which 'a' lies when the graph of the function $y = 16x^2 + 8(a+5) - 7a - 5$ is strictly above the x-axis is

Sol. $\Delta < 0 \Rightarrow 64(a+5)^2 + 64(7a+5) < 0$

$$\Rightarrow 64(a+2)(a+2) < 0$$

$$\therefore -15 < a < -2.$$



Concept Application

15. The value of 'a' for which one root of the quadratic equation $(a^2 - 5a + 3)x^2 - (3a - 1)x + 2 = 0$ is twice as large as other is

- (a) $-2/3$ (b) $1/3$ (c) $-1/3$ (d) $2/3$

EQUATIONS REDUCIBLE TO QUADRATIC EQUATIONS

1. **Reciprocal equation** of the standard form can be reduced to an equation of half its dimension.

2. **Equations of the form**

(a) $(x-a)(x-b)(x-c)(x-d) = A$ where $a < b < c < d$ can be solved by change of variable.

$$y = x - \left(\frac{a+b+c+d}{4} \right)$$

(b) $(x-a)(x-b)(x-c)(x-d) = Ax^2$ where

$$ab = cd \text{ can be solved by assumption } y = x + \frac{ab}{x}.$$

3. **For the equation of the type** $(x-a)^4 + (x-b)^4 = A$

$$\text{Substitute } y = \frac{(x-a)+(x-b)}{2}$$

IMPORTANT THEOREMS AND RESULTS

- If α is a root of the equation $f(x) = 0$, then $f(x)$ is exactly divisible by $(x - \alpha)$ and conversely, if $f(x)$ is exactly divisible by $(x - \alpha)$ then α is a root of the equation $f(x) = 0$.
- Every equation of an odd degree has atleast one real root, whose sign is opposite to that of its last term, provided that the coefficient of the first term is positive.
- Every equation of an even degree whose last term is negative has at least two real roots, one positive and one negative, provided that the coefficient of the first term is positive.

4. If an equation has no odd powers of x , then all roots of the equation are complex provided all the coefficients of the equation are having positive sign.
5. If $x = \alpha$ is root repeated m times in $f(x) = 0$,
 $(f(x) = 0$ is an n th degree equation in x) then
 $f(x) = (x - \alpha)^m g(x)$
 when $g(x)$ is a polynomial of degree $(n - m)$ and the root $x = \alpha$ is repeated $(m - 1)$ times in $f'(x) = 0$, $(m - 2)$ times in $f''(x) = 0, \dots, (m - (m - 1))$ times in $f^{m-1}(x) = 0$.
6. The condition that a quadratic function
 $f(x, y) = ax^2 + 2hxy + by^2 + 2gx + 2fy + c$ may be resolved into two linear factor is
 $\Delta = abc + 2fgh - af^2 - bg^2 - ch^2$

or $\begin{vmatrix} a & h & g \\ h & b & f \\ g & f & c \end{vmatrix}$ is 0.



Train Your Brain

Example 41: Solve $2x^4 + x^3 - 11x^2 + x + 2 = 0$

Sol. Since $x = 0$ is not a solution hence, divide by x^2
 We get,

$$2x^2 + x - 11 + \frac{1}{x} + \frac{2}{x^2} = 0$$

or $2\left(x^2 + \frac{1}{x^2}\right) + \left(x + \frac{1}{x}\right) - 11 = 0$

Let $x + \frac{1}{x} = y \Rightarrow 2(y^2 - 2) + y - 11 = 0$
 or $2y^2 + y - 15 = 0 \Rightarrow y = -3, \frac{5}{2}$

Corresponding values of x are $\frac{1}{2}, 2, \frac{-3 \pm \sqrt{5}}{2}$

Example 42: If the equations $4x^2 - 11x + 2k = 0$ and $x^2 - 3x - k = 0$ have a common root, then the value of k and common root is

(a) $0, \frac{17}{6}$ (b) $\frac{-17}{36}, \frac{17}{6}$ (c) $0, \frac{-17}{6}$ (d) $\frac{-17}{6}, 0$

Sol. The condition for the equations $ax^2 + bx + c = 0$ and $a'x^2 + b'x + c' = 0$ to have a common root α is

$$(ca' - c'a)2 = (bc' - b'c)(ab' - a'b) \text{ and } \alpha = \frac{ca' - c'a}{ab' - a'b}$$

$$4x^2 - 11x + 2k = 0 \Rightarrow x^2 - 3x - k = 0$$

$$\therefore (2k + 4k)^2 = (11k + 6k)(-12 + 11)$$

$$\therefore (6k)^2 = -17k \Rightarrow k = 0 \text{ or } k = -\frac{17}{36}$$

The common root α is given by $\alpha = \frac{6k}{-1}$

$\therefore$ If $k = 0$, then $\alpha = 0$ and if $k = -\frac{17}{36}$ then $\alpha = \frac{17}{6}$.

Hence answer is (b)

Example 43: The value of $\sqrt{7 + \sqrt{7 - \sqrt{7 + \sqrt{7 - \dots}}}}$ is
 (a) 5 (b) 4 (c) $\frac{1}{3}$ (d) 2

Sol. Let $y = \sqrt{7 + \sqrt{7 - \sqrt{7 + \sqrt{7 - \dots}}}}$

or $y = \sqrt{7 + \sqrt{7 - y}}$

or $y^2 - 7 = \sqrt{7 - y} \quad \dots(i)$

Clearly $y^2 - 7 \geq 0 \Rightarrow y^2 \geq 7$

$\Rightarrow y \geq \sqrt{7}$ or $y \leq -\sqrt{7}$

$y \leq -\sqrt{7}$ is not acceptable as $y \geq 0$ so only solution

may be in the interval $y \geq \sqrt{7}$

Since $\sqrt{7} > 2$ so $y = 2$ is not possible

Keeping $y = 3, 4$ and 5 in equation (i) we see

That only $y = 3$ satisfies it and not $y = 4$ or $y = 5$.

So only solution is $y = 3$.

Example 44: If $0 \leq x \leq \pi$, then the solution of the equation $16^{\sin^2 x} + 16^{\cos^2 x} = 10$ is given by x equal to

(a) $\frac{\pi}{6}, \frac{\pi}{3}$

(b) $\frac{\pi}{3}, \frac{\pi}{2}$

(c) $\frac{\pi}{6}, \frac{\pi}{2}$

(d) None of these

Sol. Let $16^{\sin^2 x} = y$, then $16^{\cos^2 x} = 16^{1 - \sin^2 x} = \frac{16}{y}$

Hence, $y + \frac{16}{y} = 10$

$\Rightarrow y^2 - 10y + 16 = 0$ or $y = 2, 8$

Now $16^{\sin^2 x} = 2 \Rightarrow 4 \sin^2 x = 1$

$\therefore \sin x = \pm \frac{1}{2} \Rightarrow x = \frac{\pi}{6}$

Also $16^{\sin^2 x} = 8 \Rightarrow 4 \sin^2 x = 3$

$\therefore \sin x = \pm \frac{\sqrt{3}}{2} \Rightarrow x = \frac{\pi}{3}$

Hence (a) is the correct answer.

Example 45: The root of $x^2 + 5|x| + 4 = 0$ are

Sol. Write $x^2 + 5|x| + 4 = 0$ as $|x|^2 + 5|x| + 4 = 0$

$\Rightarrow |x| = -1, |x| = -4$ (which is not possible if $x \in \mathbb{R}$)

$\therefore$ The given equation has no real roots.



Concept Application

16. If $x^4 + 4xy + 4y^2 + 4x + cy + 3$ can be written as the product of two linear factors, then $c =$

(a) 2

(b) 4

(c) 6

(d) 8

17. If $(7 + 4\sqrt{3})^{x^2-8} + (7 - 4\sqrt{3})^{x^2-8} = 14$ then the value of x is

Short Notes

Solution of Quadratic Equation & Relation Between Roots & Co-Efficients

- (a) The solutions of the quadratic equation, $ax^2 + bx + c = 0$ is given by $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$
- (b) The expression $b^2 - 4ac = D$ is called the discriminant of the quadratic equation.
- (c) If α & β are the roots of the quadratic equation $ax^2 + bx + c = 0$, then;
- $\alpha + \beta = -b/a$
 - $\alpha\beta = c/a$
 - $|\alpha - \beta| = \sqrt{D}/|a|$
- (d) Quadratic equation whose roots are α & β is $(x - \alpha)(x - \beta) = 0$ i.e., $x^2 - (\alpha + \beta)x + \alpha\beta = 0$ i.e. $x^2 - (\text{sum of roots})x + \text{product of roots} = 0$.

Nature of Roots

- (a) Consider the quadratic equation $ax^2 + bx + c = 0$ where $a, b, c \in R$ & $a \neq 0$ then;
- $D > 0 \Leftrightarrow$ roots are real & distinct (unequal).
 - $D = 0 \Leftrightarrow$ roots are real & coincident (equal).
 - $D < 0 \Leftrightarrow$ roots are imaginary.
 - If $p + iq$ is one root of a quadratic equation, then the other root must be the conjugate $p - iq$ & vice versa.
($p, q \in R$ & $i = \sqrt{-1}$).
- (b) Consider the quadratic equation $ax^2 + bx + c = 0$ where $a, b, c \in Q$ & $a \neq 0$ then;
- If D is a perfect square, then roots are rational.
 - If $\alpha = p + \sqrt{q}$ is one root in this case, (where p is rational & $\sqrt{q}$ is a surd) then other root will be $p - \sqrt{q}$. (if a, b, c are rational)

Common Roots of Two Quadratic Equations

- (a) Only one common root.

Let α be the common root of $ax^2 + bx + c = 0$ & $a'x^2 + b'x + c' = 0$ then $a\alpha^2 + b\alpha + c = 0$ & $a'\alpha^2 + b'\alpha + c' = 0$.
By Cramer's Rule

$$\frac{\alpha^2}{bc' - b'c} = \frac{\alpha}{a'c - ac'} = \frac{1}{ab' - a'b}$$

$$\text{Therefore, } \alpha = \frac{ca' - c'a}{ab' - a'b} = \frac{bc' - b'c}{a'c - ac'}$$

So the condition for a common root is $(ca' - c'a)^2 = (ab' - a'b)(bc' - b'c)$

- (b) If both roots are same then $\frac{a}{a'} = \frac{b}{b'} = \frac{c}{c'}$

Roots Under Particular Cases

Let the quadratic equation $ax^2 + bx + c = 0$ has real roots and

- If $b = 0 \Rightarrow$ roots are of equal magnitude but of opposite sign
- If $c = 0 \Rightarrow$ one root is zero other is $-b/a$
- If $a = c \Rightarrow$ roots are reciprocal to each other
- If $\begin{cases} a > 0, c < 0 \\ a < 0, c > 0 \end{cases} \Rightarrow$ roots are of opposite signs.
- If $\begin{cases} a > 0, b > 0, c > 0 \\ a < 0, b < 0, c < 0 \end{cases} \Rightarrow$ both roots are negative.
- If $\begin{cases} a > 0, b < 0, c > 0 \\ a < 0, b > 0, c < 0 \end{cases} \Rightarrow$ both roots are positive.
- If sign of $a =$ sign of $b \neq$ sign of $c \Rightarrow$ Greater root in magnitude is negative.
- If sign of $b =$ sign of $c \neq$ sign of $a \Rightarrow$ Greater root in magnitude is positive.
- If $a + b + c = 0 \Rightarrow$ one root is 1 and second root is c/a .

Maximum & Minimum Values of Quadratic Expression

Maximum & Minimum Values of expression $y = ax^2 + bx + c$

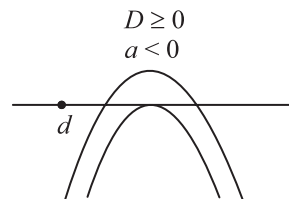
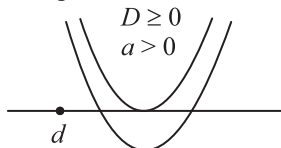
is $\frac{-D}{4a}$ which occurs at $x = -(b/2a)$ according as $a < 0$ or $a > 0$.

$$y \in \left[\frac{-D}{4a}, \infty \right) \text{ if } a > 0 \quad \& \quad y \in \left(-\infty, \frac{-D}{4a} \right] \text{ if } a < 0.$$

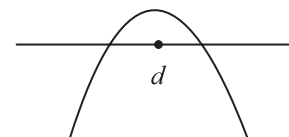
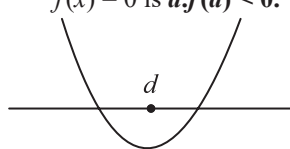
Location of roots

Let $f(x) = ax^2 + bx + c$, where $a, b, c \in R, a \neq 0$

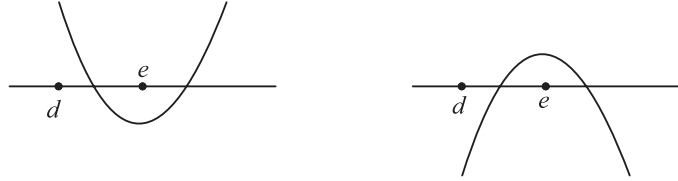
- (a) Conditions for both the roots of $f(x) = 0$ to be greater than a specified number ' d ' are $D \geq 0$; $a.f(d) > 0$ & $(-b/2a) > d$.



- (b) Conditions for the both roots of $f(x) = 0$ to lie on either side of the number ' d ' in other words the number ' d ' lies between the roots of $f(x) = 0$ is $a.f(d) < 0$.

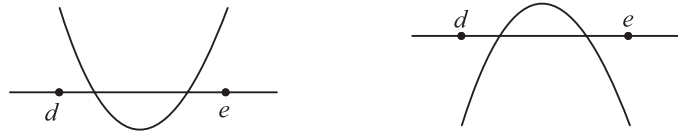


(c) Conditions for exactly one root of $f(x) = 0$ to lie in the interval (d, e) i.e., $d < x < e$ is $f(d) \cdot f(e) < 0$.



(d) Conditions that both roots of $f(x) = 0$ to be confined between the numbers d & e are (here $d < e$).

$$D \geq 0; a \cdot f(d) > 0 \text{ \& } af(e) > 0; d < (-b/2a) < e$$



General Quadratic Expression in two Variables

$f(x, y) = ax^2 + 2hxy + by^2 + 2gx + 2fy + c$ may be resolved into two linear factors if;

$$\Delta = abc + 2fgh - af^2 - bg^2 - ch^2 = 0 \text{ OR } \begin{vmatrix} a & h & g \\ h & b & f \\ g & f & c \end{vmatrix} = 0$$

Theory of Equations

If $\alpha_1, \alpha_2, \alpha_3, \dots, \alpha_n$ are the roots of the equation; $f(x) = a_0 x^n + a_1 x^{n-1} + a_2 x^{n-2} + \dots + a_{n-1} x + a_n = 0$ where $a_0, a_1, \dots, a_n$ are constants $a_0 \neq 0$ then,

$$\sum \alpha_1 = -\frac{a_1}{a_0}, \sum \alpha_1 \alpha_2 = +\frac{a_2}{a_0}, \sum \alpha_1 \alpha_2 \alpha_3 = -\frac{a_3}{a_0},$$

$$\alpha_1 \alpha_2 \alpha_3 \dots \alpha_n = (-1)^n \frac{a_n}{a_0}$$

Note:

(i) Every odd degree equation has at least one real root whose sign is opposite to that of its last term, when coefficient of highest degree term is (+)ve {If not then make it (+)ve}.

$$\text{Ex. } x^3 - x^2 + x - 1 = 0$$

(ii) Even degree polynomial whose last term is (-)ve & coefficient of highest degree term is (+)ve has atleast two real roots, one (+)ve & one (-)ve.

(iii) If equation contains only even power of x & all coefficient are (+)ve, then all roots are imaginary.

Solved Examples

1. $\sqrt{x+3-4\sqrt{x-1}} + \sqrt{x+8-6\sqrt{x-1}} = 1$. Solve for x .

- (a) $\{5, 10\}$ (b) $[1, \infty)$
(c) $[5, 10]$ (d) $(5, 10)$

Sol. (c) Put $x - 1 = t^2$ in the given equation, we get

$$\sqrt{t^2 + 1 + 3 - 4t} + \sqrt{t^2 + 1 + 8 - 6t} = 1$$

$$\Rightarrow |t - 2| + |t - 3| = 1$$

$$\Rightarrow t \in [2, 3]$$

$$\Rightarrow x \in [5, 10]$$

2. If α & β are the two distinct roots of $x^2 + 2(K - 3)x + 9 = 0$, then find the values of K such that $\alpha, \beta \in (-6, 1)$.

Sol. $x^2 + 2(k - 3)x + 9 = 0$... (i)

Roots α, β of equation (i) are distinct and lies between -6 and 1

$$D > 0$$

$$\Rightarrow 4(k - 3)^2 - 36 > 0$$

$$\Rightarrow k(k - 6) > 0$$

$$\Rightarrow k \in (-\infty, 0) \cup (6, \infty)$$

$$f(1) > 0$$

$$\Rightarrow 1 + 2(k - 3) + 9 > 0$$

... (ii)

$$\Rightarrow k \in (-2, \infty)$$

$$f(-6) > 0$$

$$\Rightarrow 36 - 12(k - 3) + 9 > 0$$

$$\Rightarrow 4k - 27 < 0 \Rightarrow k \in \left(-\infty, \frac{27}{4}\right)$$

$$-6 < -\frac{b}{2a} < 1 \Rightarrow -6 < \frac{-2(k - 3)}{2} < 1$$

$$\Rightarrow -1 < k - 3 < 6 \Rightarrow 2 < k < 9$$

$$(ii) \cap (iii) \cap (iv) \cap (v) \Rightarrow k \in \left(6, \frac{27}{4}\right)$$

3. If α is a root of the equation $4x^2 + 2x - 1 = 0$ then the other root must be

$$(a) 3\alpha^3 - 4\alpha$$

$$(b) 4\alpha^3 - 3\alpha$$

$$(c) 2\alpha^2 - 1$$

$$(d) \text{ all above}$$

Sol. (c) $4x^2 + 2x - 1 = 0$

$$x = \frac{-2 \pm \sqrt{4 + 16}}{8}$$

$$x = \frac{-2 \pm 2\sqrt{5}}{8} = \frac{-1 \pm \sqrt{5}}{4}$$

$$\text{If } \alpha = \frac{\sqrt{5}-1}{4} = \sin 18^\circ$$

$$\beta = -\left[\frac{\sqrt{5}+1}{4}\right] = -\cos 36^\circ$$

$$\beta = -\cos 36^\circ = -\sin [90^\circ - 36^\circ] = -\sin 54^\circ = 4 \sin^3 18^\circ - 3 \sin 18^\circ = 4\alpha^3 - 3\alpha$$

$$\beta = -\cos 36^\circ = 2 \cos^2 18^\circ - 1 = 2\alpha^2 - 1$$

4. Solve the equation $x(x+1)^2(x+2) = 72$.

Sol. We have $x(x+1)^2(x+2) = 72$

$$\Rightarrow (x+0)(x+1)(x+1)(x+2) = 72 \quad \dots(i)$$

$$(i) [(x+0)(x+2)][(x+1)(x+1)] = 72 \quad (\because 0+2=1+1)$$

$$\Rightarrow (x^2+2x)(x^2+2x+1) = 72 \quad \dots(ii)$$

$$\text{Let } y = x^2 + 2x$$

$$(ii) y(y+1) = 72 \Rightarrow y^2 + y - 72 = 0$$

$$\Rightarrow y = \frac{-1 \pm \sqrt{1+288}}{2} = \frac{-1 \pm 17}{2} = 8, -9$$

$$\therefore \text{Either } y = 8 \Rightarrow x^2 + 2x = 8 \Rightarrow x^2 + 2x - 8 = 0$$

$$\Rightarrow x = \frac{-2 \pm \sqrt{4+32}}{2} = \frac{-2 \pm 6}{2} = 2, -4$$

$$\text{or } y = -9 \Rightarrow x^2 + 2x = -9 \Rightarrow x^2 + 2x + 9 = 0$$

$$\Rightarrow x = \frac{-2 \pm \sqrt{4-36}}{2} = \frac{-2 \pm 4\sqrt{2}i}{2} = -1 \pm 2\sqrt{2}i$$

5. If the roots of the quadratic equation $x^2 + mx + n = 0$ are $\tan 30^\circ$ and $\tan 15^\circ$, respectively, then find the value of $2 + n - m$.

Sol. The equation $x^2 + mx + n = 0$ has roots $\tan 30^\circ$ and $\tan 15^\circ$.

Therefore,

$$\tan 30^\circ + \tan 15^\circ = -m \quad \dots(i)$$

$$\tan 30^\circ \tan 15^\circ = n \quad \dots(ii)$$

$$\text{Now, } \tan 45^\circ = \tan (30^\circ + 15^\circ) \Rightarrow 1 = \frac{\tan 30^\circ + \tan 15^\circ}{1 - \tan 30^\circ \tan 15^\circ}$$

$$\tan 30^\circ \tan 15^\circ = n \quad \dots(iii)$$

$$\text{Now, } \tan 45^\circ = \tan (30^\circ + 15^\circ) \Rightarrow 1 = \frac{\tan 30^\circ + \tan 15^\circ}{1 - \tan 30^\circ \tan 15^\circ}$$

$$\Rightarrow 1 = \frac{-m}{1-n} \quad [\text{Using (i) and (ii)}]$$

$$\Rightarrow 1 - n = -m \Rightarrow n - m = 1$$

$$\Rightarrow 2 + n - m = 3$$

6. If the difference between the roots of the equation $x^2 + ax + 1 = 0$ is less than $\sqrt{5}$, then find the set of possible values of a .

Sol. If α, β are roots of $x^2 + ax + 1 = 0$, then

$$|\alpha - \beta| < \sqrt{5}$$

$$\Rightarrow \left| \sqrt{(\alpha - \beta)^2 - 4\alpha\beta} \right| < \sqrt{5}$$

$$\Rightarrow \left| \sqrt{a^2 - 4} \right| < \sqrt{5}$$

$$\Rightarrow a^2 - 4 < 5$$

$$\Rightarrow a^2 - 9 < 0$$

$$\therefore a \in (-3, 3)$$

7. Find the values of a for which one root of equation $(a-5)x^2 - 2ax + a - 4 = 0$ is smaller than 1 and the other greater than 2.

Sol. Let $f(x) = (a-5)x^2 - 2ax + a - 4$ ($a \neq 5$) as 1 and 2 lie between the roots of $f(x) = 0$

$$\text{Consider } (a-5)f(1) < 0: (a-5)(a-5-2a+a-4) < 0$$

$$\Rightarrow (a-5)(-9) < 0$$

$$\Rightarrow a - 5 > 0$$

$$\Rightarrow a \in (5, \infty) \quad \dots(i)$$

$$\text{Consider } (a-5)f(2) < 0: (a-5)(4(a-5) - 4a + a - 4) < 0$$

$$\Rightarrow (a-5)(a-24) < 0$$

$$\Rightarrow a \in (5, 24) \quad \dots(ii)$$

Hence, the values of a satisfying (i) and (ii) at the same time are $a \in (5, 24)$

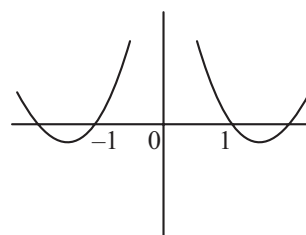
8. Find all values of k for which the inequality, $2x^2 - 4k^2x - k^2 + 1 > 0$ is valid for all real x which do not exceed unity in the absolute value.

Sol. Case 1: When $d < 0$ which gives the result

$$\text{Case 2: } f(1) \geq 0; D \geq 0; -\frac{b}{2a} > 1$$

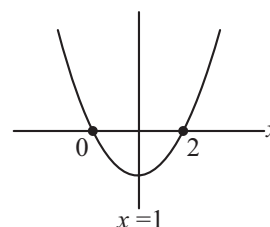
$$\text{or } f(-1) \geq 0; d \geq 0; -\frac{b}{2a} < -1$$

$$\text{No solution in case-II; } -\frac{1}{\sqrt{2}} < k < \frac{1}{\sqrt{2}}$$



9. Let $f(x) = ax^2 + bx + c$ be a polynomial. If $f(1-x) = f(1+x)$, $f(2) = 0$ and $f(3) = 6$, then find the value of a, b and c .

Sol. $f(1-x) = f(1+x) \Rightarrow$ the graph of the function $y = f(x)$ will be symmetric about the line $x = 1$



$$f(2) = 0 \Rightarrow f(0) = 0$$

$$f(x) = ax(x-2)$$

$$\text{Now } f(3) = 6 \Rightarrow a = 2$$

$$f(x) = 2x^2 - 4x$$

10. Find all values of a for which the equation $2x^2 - 2(2a+1)x + a(a+1) = 0$ has two roots, one of which is greater than a and the other is smaller than a .

Sol. Here coefficient of x^2 is positive



$$\text{Let } f(x) = 2x^2 - 2(2a+1)x + a(a+1)$$

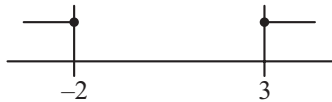
$$\therefore f(x) = 2a^2 - 2(2a+1)a + a(a+1) = -a^2 - a < 0$$

$$\Rightarrow -a(a+1) < 0 \Rightarrow a(a+1) > 0$$

$$a \in (-\infty, -1) \cup (0, \infty)$$

11. If x is real and $4y^2 + 4xy + x + 6 = 0$, then find the complete set of values of x for which y is real.

Sol. $4y^2 + 4xy + x + 6 = 0 \quad y \in R$



$$\therefore D \geq 0$$

$$16x^2 - 16(x+6) \geq 0$$

$$\Rightarrow x^2 - x - 6 \geq 0 \Rightarrow (x-3)(x+2) \geq 0$$

$$\therefore x \in (-\infty, -2] \cup [3, \infty)$$

12. If the equation $x^4 + x^3 - 2ax^2 - ax + a^2 = 0$ has 4 real roots then find the least possible value of ' a '.

Sol. To solve this type of equation try to make it a quadratic equation in ' a '.

$$a^2 - a(2x^2 + x) + x^4 + x^3 = 0$$

$$\Rightarrow a^2 - a(x^2 + x^2 + x) + x^4 + x^3 = 0$$

$$\Rightarrow (x^2 - a)(x^2 + x - a) = 0$$

If all roots are real then both factors must have real roots.

$$x^2 - a = 0 \text{ and } x^2 + x - a = 0$$

$$a \geq 0 \text{ and } 1 + 4a \geq 0 \Rightarrow a \geq -\frac{1}{4}$$

$\Rightarrow$ the least possible value of ' a ' is 0.

13. If $Q_1(x) = x^2 + (k-29)x - k$ and $Q_2(x) = 2x^2 + (2k-43)x + k$ both are factors of a cubic polynomial $P(x)$, then the largest value of k is

(a) 0 (b) 33 (c) 23 (d) 30

Sol. Two quadratic polynomials can be a factor of cubic polynomial only when they have atleast one root common

$$\Rightarrow x^2 + (b-2a)x - k = 0 \quad \dots(i)$$

$$\text{and } 2x^2 + (2k-43)x + k = 0 \quad \dots(ii)$$

Must have a common roots

Multiple equation (i) by (ii) and subtracting, we get

$$15x + 3k = 0 \Rightarrow x = \frac{-k}{5} \text{ is the common root and it must}$$

satisfy equation (i)

$$\Rightarrow \frac{k^2}{25} + (k-29)\left(-\frac{k}{5}\right) - k = 0$$

$$\Rightarrow \left(-\frac{k}{5}\right)\left[-\frac{k}{5} + k - 29 + 5\right] = 0 \Rightarrow k = 0 \text{ or } k = 30$$



Exercise-1 (Topicwise)

QUADRATIC EQUATION & NATURE OF ROOTS

- If $a + b + c = 0$, and $a, b, c \in \mathbb{R}$, then the roots of the equation $4ax^2 + 3bx + 2c = 0$ are
 - Equal
 - Imaginary
 - Real
 - Both (a) and (b)
- The roots of the given equation $2(a^2 + b^2)x^2 + 2(a + b)x + 1 = 0$ are
[Where $a \neq b$]
 - Rational
 - Irrational
 - Real
 - Imaginary
- If $a, b, c \in \mathbb{Q}$, then roots of the equation $(b + c - 2a)x^2 + (c + a - 2b)x + (a + b - 2c) = 0$ are
 - Rational
 - Non-real
 - Irrational
 - Equal
- The value of k for which the quadratic equation, $kx^2 + 1 = kx + 3x - 11x^2$ has real and equal roots are
 - 11, -3
 - 5, 7
 - 5, -7
 - 7, 25
- $x^2 + x + 1 + 2k(x^2 - x - 1) = 0$ is a perfect square for how many values of k
 - 2
 - 0
 - 1
 - 3
- If $x^2 - 3x + 2$ be a factor of $x^4 - px^2 + q$, then $(pq) =$
 - (3, 4)
 - (4, 5)
 - (4, 3)
 - (5, 4)
- If a, b, c are integers and $b^2 = 4(ac + 5d^2)$, $d \in \mathbb{N}$, then roots of the equation $ax^2 + bx + c = 0$ are
 - Irrational
 - Rational & different
 - Complex conjugate
 - Rational & equal
- Consider the equation $x^2 + 2x - n = 0$, where $n \in \mathbb{N}$ and $n \in [5, 100]$. Total number of different values of 'n' so that the given equation has integral roots, is
 - 4
 - 6
 - 8
 - 3
- The entire graph of the expression $y = x^2 + kx - x + 9$ is strictly above the x-axis if and only if
 - $k < 7$
 - $-5 < k < 7$
 - $k > -5$
 - None
- If $a, b \in \mathbb{R}$, $a \neq 0$ and the quadratic equation $ax^2 - bx + 1 = 0$ has imaginary roots then $a + b + 1$ is:
 - positive
 - negative
 - zero
 - depends on the sign of b
- If a and b are the non-zero distinct roots of $x^2 + ax + b = 0$, then the least value of $x^2 + ax + b$ is
 - $\frac{3}{2}$
 - $\frac{9}{4}$
 - $-\frac{9}{4}$
 - 1
- If both roots of the quadratic equation $(2 - x)(x + 1) = p$ are distinct & positive, then p must lie in the interval
 - $(2, \infty)$
 - $(2, 9/4)$
 - $(-\infty, -2)$
 - $(-\infty, \infty)$
- If $(1 - p)$ is root of quadratic equation $x^2 + px + (1 - p) = 0$, then its roots are
 - 0, 1
 - 1, 1
 - 0, -1
 - 1, 2
- If $a + b + c = 0$ then the equation $3ax^2 + 2bx + c = 0$ has
 - Imaginary roots
 - Real and equal roots
 - Real and different roots
 - Rational roots
- If a, b, c are in A.P. the roots of the equation $ax^2 + 2bx + c = 0$ are
 - Real and distinct
 - Real and equal
 - Real
 - Imaginary
- If the roots of the equation $bx^2 + cx + a = 0$ be imaginary then for all real values of x , the expression $3b^2x^2 + 6bcx + 2c^2$ is
 - Greater than $4ab$
 - Less than $4ab$
 - Greater than $-4ab$
 - Less than $-4ab$

Sum and Product of Roots

- If one root of $5x^2 + 13x + k = 0$ is reciprocal of the other, then $k =$
 - 0
 - 5
 - $1/6$
 - 6
- If the sum of the roots of the equation $ax^2 + bx + c = 0$ be equal to the sum of their squares, then
 - $a(a + b) = 2bc$
 - $c(a + c) = 2ab$
 - $b(a + b) = 2ac$
 - $b(a + b) = a$
- If α, β be the roots of the equation $x^2 - 2x + 3 = 0$, then the equation whose roots are $\frac{1}{\alpha^2}$ and $\frac{1}{\beta^2}$ is
 - $x^2 + 2x + 1 = 0$
 - $9x^2 + 2x + 1 = 0$
 - $9x^2 - 2x + 1 = 0$
 - $9x^2 + 2x - 1 = 0$
- If α and β are the roots of the equation $x^2 - 4x + 1 = 0$ the value of $\alpha^3 + \beta^3$ is
 - 76
 - 52
 - 52
 - 76

21. If a, b are the roots of quadratic equation $x^2 + px + q = 0$ and g, d are the roots of $x^2 + px - r = 0$, then $(a - g) \cdot (a - d)$ is equal to:
- (a) $q + r$ (b) $q - r$
(c) $-(q + r)$ (d) $-(p + q + r)$
22. Two real numbers a & b are such that $a + b = 3$ & $|a - b| = 4$, then a & b are the roots of the quadratic equation:
- (a) $4x^2 - 12x - 7 = 0$ (b) $4x^2 - 12x + 7 = 0$
(c) $4x^2 - 12x + 25 = 0$ (d) None of these
23. Let conditions C_1 and C_2 be defined as follows:
 $C_1 : b^2 - 4ac \geq 0$, $C_2 : a, -b, c$ are of same sign. The roots of $ax^2 + bx + c = 0$ are real and positive, if
- (a) both C_1 and C_2 are satisfied
(b) only C_2 is satisfied
(c) only C_1 is satisfied
(d) None of these
24. If α, β are roots of the equation $ax^2 + bx + c = 0$, then the value of $\alpha^3 + \beta^3$ is
- (a) $\frac{3abc + b^3}{a}$ (b) $\frac{a^3 + b^3}{3abc}$
(c) $\frac{3abc - b^3}{a^3}$ (d) $\frac{-(3abc + b^3)}{a^3}$
25. If $\alpha + \beta = -2$ and $\alpha^2 + \beta^2 = -56$ then the quadratic equation whose roots are α, β is
- (a) $x^2 + 2x - 16 = 0$ (b) $x^2 + 2x - 15 = 0$
(c) $x^2 + 2x - 12 = 0$ (d) $x^2 + 2x - 8 = 0$
26. $\tan 22^\circ$ and $\tan 23^\circ$ are roots of $x^2 + ax + b = 0$ then
- (a) $a + b + 1 = 0$ (b) $a - b + 1 = 0$
(c) $b - a + 1 = 0$ (d) $a + b = 1$
27. Let α, β be the roots of the equation $(x - a)(x - b) = c$, $c \neq 0$ then the roots of the equation $(x - \alpha)(x - \beta) + c = 0$ are
- (a) a, c (b) b, c
(c) a, b (d) $a + b, b + c$
28. Let α, β are the roots of the equation $x^2 - 15x + 1 = 0$, then the value of $\left(\frac{1}{\alpha} - 15\right)^{-2} + \left(\frac{1}{\beta} - 15\right)^{-2}$ is
- (a) 225 (b) 900 (c) 223 (d) 0
29. The quadratic equation whose roots are the x and y intercepts of the line passing through $(1, 1)$ and making a triangle of area A with the axes, may be
- (a) $x^2 + Ax + 2A = 0$ (b) $x^2 - 2Ax + 2A = 0$
(c) $x^2 - Ax + 2A = 0$ (d) $x^2 - 2Ax - 2A = 0$
30. If one root of $x^2 + px + 12 = 0$ is 4 while the equation $x^2 + px + q = 0$ has equal roots then $q =$
- (a) -7 (b) 4
(c) 42 (d) $\frac{49}{4}$
31. If $k > 0$ and the product of the roots of the equation $x^2 - 3kx + 2e^{2\log k} - 1 = 0$ is 7 then the sum of the roots is
- (a) 2 (b) 4 (c) 6 (d) 8
32. Two students while solving a quadratic equation in x , one copied the constant term incorrectly and got the roots as 3 and 2. The other copied the constant terms and coefficient of x^2 as -6 and 1 respectively. The correct roots are
- (a) 3, -2 (b) -3, 2
(c) -6, -1 (d) 6, -1
33. Let A, G and H be the A.M., G.M. and H.M. of two positive number a and b . The quadratic equation whose roots are A and H is
- (a) $Ax^2 - (A^2 + G^2)x + AG^2 = 0$
(b) $Ax^2 - (A^2 + G^2)x + AH^2 = 0$
(c) $Hx^2 - (H^2 + G^2)x + HG^2 = 0$
(d) $Gx^2 - (H^2 + G^2)x + GH^2 = 0$
34. If one root of the equation $ax^2 + bx + c = 0$ is the square of the other, then
- (a) $b^2 + ac^2 + a^2c = 3abc$
(b) $b^3 + ac^2 + a^2c = 3abc$
(c) $b^2 + ac^2 + a^2c + 3abc = 0$
(d) $b^3 + ac^2 + a^2c + 3abc = 0$
35. The roots of the equation $ax^2 + bx + c = 0$, $a \in R^+$, are two consecutive odd positive integers. Then
- (a) $|b| \leq 4a$ (b) $|b| \geq 4a$
(c) $|b| \geq 2a$ (d) $|b| \leq a$
36. The number of real solution of the equation $(9/10)^x = -3 + x - x^2$ is
- (a) 2 (b) 0
(c) 1 (d) 3
37. If the roots of $x^2 - bx + c = 0$ are two consecutive integers then $b^2 - 4c =$
- (a) 0 (b) 1 (c) 2 (d) 3
38. If the roots of the quadratic equation $x^2 + px + q$ are $\tan 30^\circ$ and $\tan 15^\circ$ respectively then the value of $2 + q - p$ is
- (a) 1 (b) 2
(c) 3 (d) 0

COMMON ROOTS

39. If both the roots of $k(6x^2 + 3) + rx + 2x^2 - 1 = 0$ and $6k(2x^2 + 1) + px + 4x^2 - 2 = 0$ are common, then $2r - p$ is equal to
- (a) -1 (b) 0
(c) 1 (d) 2
40. If a, b, p, q are non-zero real numbers, then the equations $2a^2x^2 - 2abx + b^2 = 0$ and $p^2x^2 + 2pqx + q^2 = 0$ have:
- (a) no common root
(b) one common root if $2a^2 + b^2 = p^2 + q^2$
(c) two common roots if $3pq = 2ab$
(d) two common roots if $3qb = 2ap$

41. The equations $x^2 + 3x + 5 = 0$ and $ax^2 + bx + c = 0$ have a common root. If $a, b, c \in \mathbb{N}$ then the least possible values of $a + b + c$ is equal to
 (a) 3 (b) 6 (c) 9 (d) 12
42. If the equation $ax^2 + 2bx + 3c = 0$ and $3x^2 + 8x + 15 = 0$ have a common root where a, b, c are the length of the sides of $\triangle ABC$ then $\sin^2 A + \sin^2 B + \sin^2 C$ is equal to
 (a) 1 (b) $\frac{3}{2}$
 (c) $\sqrt{2}$ (d) 2

Theory of Equation and Identity, Inequalities

43. If x is real and satisfies $x + 2 > \sqrt{x + 4}$, then
 (a) $x < -2$ (b) $x > 0$
 (c) $-3 < x < 0$ (d) $-3 < x < 4$
44. The set of all real numbers x for which $x^2 - |x + 2| + x > 0$ is
 (a) $(-\infty, -2) \cup (2, \infty)$ (b) $(-\infty, -\sqrt{2}) \cup (\sqrt{2}, \infty)$
 (c) $(-\infty, -1) \cup (1, \infty)$ (d) $(\sqrt{2}, \infty)$
45. Number of values of ' p ' for which the equation $(p^2 - 3p + 2)x^2 - (p^2 - 5p + 4)x + p - p^2 = 0$ possess more than two roots, is:
 (a) 0 (b) 1
 (c) 2 (d) None
46. The complete set of values of ' x ' which satisfy the in equations : $5x + 2 < 3x + 8$ and $\frac{x + 2}{x - 1} < 4$ is
 (a) $(-\infty, 1)$ (b) $(2, 3)$
 (c) $(-\infty, 3)$ (d) $(-\infty, 1) \cup (2, 3)$
47. If the inequality $(m - 2)x^2 + 8x + m + 4 > 0$ is satisfied for all $x \in \mathbb{R}$, then the least integral ' m ' is:
 (a) 4 (b) 5
 (c) 6 (d) none
48. The complete set of real ' x ' satisfying $\|x - 1| - 1| \leq 1$ is:
 (a) $[0, 2]$ (b) $[-1, 3]$
 (c) $[-1, 1]$ (d) $[1, 3]$
49. If $\log_{1/3} \frac{3x - 1}{x + 2}$ is less than unity, then ' x ' must lie in the interval:
 (a) $(-\infty, -2) \cup (5/8, \infty)$ (b) $(-2, 5/8)$
 (c) $(-\infty, -2) \cup (1/3, 5/8)$ (d) $(-2, 1/3)$
50. Solution set of the inequality $2 - \log_2(x^2 + 3x) \geq 0$ is:
 (a) $[-4, 1]$ (b) $[-4, -3] \cup (0, 1]$
 (c) $(-\infty, -3) \cup (1, \infty)$ (d) $(-\infty, -4) \cup [1, \infty)$
51. The set of all the solutions of the inequality $\log_{1-x}(x - 2) \geq 0$ is
 (a) $(-\infty, 0)$ (b) $(2, \infty)$
 (c) $(-\infty, 1)$ (d) ϕ

52. If $\log_{0.5} \log_5(x^2 - 4) > \log_{0.5} 1$, then ' x ' lies in the interval
 (a) $(-3, -\sqrt{5}) \cup (\sqrt{5}, 3)$ (b) $(-3, -\sqrt{5}) \cup (\sqrt{5}, 3\sqrt{5})$
 (c) $(\sqrt{5}, 3\sqrt{5})$ (d) ϕ
53. If $\frac{6x^2 - 5x - 3}{x^2 - 2x + 6} \leq 4$, then least and the highest values of $4x^2$ are
 (a) 0 & 81 (b) 9 & 81
 (c) 36 & 81 (d) None of these
54. If two roots of the equation $x^3 - px^2 + qx - r = 0$ are equal in magnitude but opposite in sign, then:
 (a) $pr = q$ (b) $qr = p$
 (c) $pq = r$ (d) None
55. If α, β & γ are the roots of the equation $x^3 - x - 1 = 0$ then, $\frac{1 + \alpha}{1 - \alpha} + \frac{1 + \beta}{1 - \beta} + \frac{1 + \gamma}{1 - \gamma}$ has the value equal to:
 (a) zero (b) -1
 (c) -7 (d) 1
56. For what value of a and b the equation $x^4 - 4x^3 + ax^2 + bx + 1 = 0$ has four real positive roots?
 (a) $(-6, -4)$ (b) $(-6, 5)$
 (c) $(-6, 4)$ (d) $(6, -4)$
57. If the difference between the roots of the equation $x^2 + ax + 1 = 0$ is less than $\sqrt{5}$, then the set of possible values of a is
 (a) $(-3, 3)$ (b) $(-3, \infty)$
 (c) $(3, \infty)$ (d) $(\infty, -3)$
58. The greatest negative integer satisfying $x^2 + 4x - 77 < 0$ and $x^2 > 4$ is
 (a) -1 (b) -2
 (c) -3 (d) -10
59. The solution set contained in $\mathbb{R}$ of the in equation $3^x + 3^{1-x} - 4 < 0$ is
 (a) $(1, 3)$ (b) $(0, 1)$
 (c) $(1, 2)$ (d) $(0, 2)$
60. If the equation $ax^2 + 2bx - 3c = 0$ has non real roots and $\frac{3c}{4} < a + b$ then c is always
 (a) < 0 (b) > 0 (c) ≥ 0 (d) zero
61. The equation $(\cos p - 1)x^2 + (\cos p)x + (\sin p) = 0$ in the variable x has real roots. Then p can take any value in the interval
 (a) $(0, 2\pi)$ (b) $(-\pi, 0)$
 (c) $(-\pi/2, \pi/2)$ (d) $(0, \pi)$
62. If a and b ($\neq 0$) are the root of the equation $x^2 + ax + b = 0$ then the least value of $x^2 + ax + b$ ($x \in \mathbb{R}$) is
 (a) $\frac{9}{4}$ (b) $-\frac{9}{4}$
 (c) $-\frac{1}{4}$ (d) $\frac{1}{4}$

63. If $x^2 + \frac{1}{x^2} = A$ and $x - \frac{1}{x} = B$ ($|x| > 1$) then least value of $\frac{A}{B}$ is
 (a) 2 (b) $\sqrt{2}$
 (c) $-\sqrt{2}$ (d) $-2\sqrt{2}$
64. If α, β are the roots of $x^2 - (a-2)x - (a+1) = 0$ where 'a' is a variable then the least value of $\alpha^2 + \beta^2$ is
 (a) 2 (b) 3 (c) 5 (d) 7
65. The minimum value of $|x| + \left|x + \frac{1}{2}\right| + |x-3| + \left|x - \frac{5}{2}\right|$ is
 (a) 2 (b) 4 (c) 6 (d) 8
66. The real roots of the equation $|x^2 + 4x + 3| + 2x + 5 = 0$ are
 (a) $4, -1 + \sqrt{3}$ (b) $-4, -1 - \sqrt{3}$
 (c) $-6, -1$ (d) $6, -1$
67. If x satisfies $|x-1| + |x-2| + |x-3| \geq 6$ then
 (a) $0 \leq x \leq 4$ (b) $x \leq -2$ (or) $x \geq 4$
 (c) $x \leq 0$ (or) $x \geq 4$ (d) R

Max and Min Value, Factorization

68. If x is real, then the maximum and minimum values of the expression $\frac{x^2 - 3x + 4}{x^2 + 3x + 4}$ will be
 (a) 2, 1 (b) $5, \frac{1}{5}$
 (c) $7, \frac{1}{7}$ (d) 2, 7
69. The smallest value of $x^2 - 3x + 3$ in the interval $(-3, 3/2)$ is
 (a) $3/4$ (b) 5 (c) -15 (d) -20
70. If 'x' is real and $k = \frac{x^2 - x + 1}{x^2 + x + 1}$, then:
 (a) $\frac{1}{3} \leq k \leq 3$ (b) $k \geq 5$
 (c) $k \leq 0$ (d) None of these
71. Consider $y = \frac{2x}{1+x^2}$, where x is real, then the range of expression $y^2 + y - 2$ is
 (a) $[-1, 1]$ (b) $[0, 1]$
 (c) $[-9/4, 0]$ (d) $[-9/4, 1]$
72. The values of x and y besides y can satisfy the equation $(x, y \in \text{real numbers}) x^2 - xy + y^2 - 4x - 4y + 16 = 0$
 (a) 2, 2 (b) 4, 4
 (c) 3, 3 (d) None of these
73. If x is real, then $\frac{x^2 - x + c}{x^2 + x + 2c}$ can take all real values if
 (a) $c \in [0, 6]$ (b) $c \in [-6, 0]$
 (c) $c \in (-\infty, -6) \cup (0, \infty)$ (d) $c \in (-6, 0)$

LOCATION OF ROOTS

74. The real values of 'a' for which the quadratic equation $2x^2 - (a^3 + 8a - 1)x + a^2 - 4a = 0$ possesses roots of opposite sign is given by:
 (a) $a > 5$ (b) $0 < a < 4$ (c) $a > 0$ (d) $a > 7$
75. If a, b are the roots of the quadratic equation $x^2 - 2p(x-4) - 15 = 0$, then the set of values of 'p' for which one root is less than 1 & the other root is greater than 2 is:
 (a) $(7/3, \infty)$ (b) $(-\infty, 7/3)$
 (c) $x \in R$ (d) None of these
76. The values of k, for which the equation $x^2 + 2(k-1)x + k + 5 = 0$ possess atleast one positive root, are
 (a) $[4, \infty)$ (b) $(-\infty, -1] \cup [4, \infty)$
 (c) $[-1, 4]$ (d) $(-\infty, -1]$
77. The set of values of p for which 6 lies between the roots of the equation $x^2 + 2(p-3)x + 9 = 0$ is
 (a) $\left(\frac{3}{4}, \infty\right)$ (b) $\left(-\infty, -\frac{3}{4}\right)$
 (c) $\left(-\frac{3}{4}, \frac{3}{4}\right)$ (d) $\left[-\frac{3}{4}, \frac{3}{4}\right]$
78. All the values of m for which both roots of the equation $x^2 - 2mx + m^2 - 1 = 0$ are greater than 2 but less than 4, lie in the interval
 (a) $-1 < m < 3$ (b) $1 < m < 4$
 (c) $-2 < m < 0$ (d) $m > 3$
79. If $a \in R$ and the roots of $x^2 - 2x - a^2 + 1 = 0$ lies between the roots of $x^2 - 2(a+1)x + a(a-1) = 0$ then 'a' belongs to
 (a) $(1, \infty)$ (b) $\left(-\frac{1}{4}, 1\right)$ (c) $(-\infty, 0)$ (d) $\left(-\infty, -\frac{1}{4}\right)$
80. Consider the equation $(1+a+b)^2 = 3(1+a^2+b^2)$, where a, b are real numbers. Then
 (a) There is no solution pair (a, b).
 (b) There are infinitely many solution pairs (a, b).
 (c) There are exactly two solution pairs (a, b).
 (d) There is exactly one solution pair (a, b).
81. Let S be the set of all ordered pairs (x, y) of positive integers satisfying the condition $x^2 - y^2 = 12345678$. Then
 (a) S is an infinite set
 (b) S is the empty set
 (c) S has exactly one element
 (d) S is a finite set and has at least two elements
82. Consider $f(x) = 4x^4 - 24x^3 + 31x^2 + 6x - 8$ be a polynomial function and $\alpha, \beta, \gamma, \sigma$ are the roots ($\alpha < \beta < \gamma < \sigma$). So

$$\sigma^p + \frac{1}{\sigma^\alpha} + \alpha^\gamma + \gamma^\sigma$$
 (a) 36 (b) 18 (c) 9 (d) 10

83. Solve $\left(\frac{x-1}{x+2}\right)^3 = \frac{x-4}{x+5}$.

- (a) $-\frac{1}{2}$ (b) $\frac{1}{2}$ (c) $-\frac{1}{4}$ (d) $\frac{1}{4}$

84. If α, β, γ are the roots of the equation, $x^3 + px^2 + qx + r = 0$, then find the value of $\left(\alpha - \frac{1}{\beta\gamma}\right)\left(\beta - \frac{1}{\gamma\alpha}\right)\left(\gamma - \frac{1}{\alpha\beta}\right)$.

- (a) $-\frac{(r-1)^3}{r^2}$ (b) $-\frac{(r-2)^3}{r^2}$
(c) $-\frac{(r+1)^3}{r^2}$ (d) $-\frac{(r-2)^3}{(r-1)}$

85. Solve the equation $x^3 - 13x^2 + 15x + 189 = 0$, if one root exceeds the other by 2.

- (a) 7, 9, -3 (b) 2, -5, -4 (c) 1, 2, 6 (d) -1, -8, 9

86. If the inequality $\frac{mx^2 + 3x + 4}{x^2 + 2x + 2} < 5$ is satisfied for all $x \in R$,

then find the values of m .

- (a) $1 < m < 5$ (b) $-1 < m < 5$
(c) $1 < m < 6$ (d) $m < 71/24$

87. Sum of all the roots of the equation $x^2 - 2x + |x-1| - 5 = 0$ is

- (a) 0 (b) 2 (c) 1 (d) 5

88. If α and β are the distinct roots of the equation $x^2 + 3^{1/4}x + 3^{1/2} = 0$ then the value of $\alpha^{96}(\alpha^{12} - 1) + \beta^{96}(\beta^{12} - 1)$ is equal to

- (a) 56×3^{24} (b) 52×3^{24}
(c) 56×3^{25} (d) 28×3^{25}

89. For what integral k is the inequality $x^2 - 2(4k-1)x + 15k^2 - 2k - 7 > 0$ valid for any real x ?

- (a) 2 (b) 3 (c) 4 (d) 5

90. Number of real roots of the equation $x^4 - 4x - 1 = 0$ has

- (a) atmost two positive real root
(b) atmost one negative real root
(c) atmost two real roots
(d) 4 real roots

Exercise-2 (Learning Plus)

1. If $a > 0, b > 0, c > 0$ then both the roots of the equation $ax^2 + bx + c = 0$

- (a) Are real and negative
(b) Have negative real parts
(c) Are rational numbers
(d) Both (a) and (c)

2. If α, β are the roots of the equation $ax^2 + bx + c = 0$ then the equation whose roots are $\alpha + \frac{1}{\beta}$ and $\beta + \frac{1}{\alpha}$, is

- (a) $acx^2 + (a+c)bx + (a+c)^2 = 0$
(b) $abx^2 + (a+c)bx + (a+c)^2 = 0$
(c) $acx^2 + (a+b)cx + (a+c)^2 = 0$
(d) $acx^3 + (a+c)bx + (a+c)^3$

3. $x^2 - 11x + a$ and $x^2 - 14x + 2a$ will have a common factor, if $a =$

- (a) 24 (b) 0, 24
(c) 3, 24 (d) 0, 3

4. If α, β, γ are the roots of the equation $x^3 + 4x + 1 = 0$, then $(\alpha + \beta)^{-1} + (\beta + \gamma)^{-1} + (\gamma + \alpha)^{-1}$

- (a) 2 (b) 3 (c) 4 (d) 5

5. If the roots of $x^2 + x + a = 0$ exceed a , then

- (a) $2 < a < 3$ (b) $a > 3$
(c) $-3 < a < 3$ (d) $a < -2$

6. The value of p for which both the roots of the equation $4x^2 - 20px + (25p^2 + 15p - 66) = 0$ are less than 2, lies in

- (a) $(4/5, 2)$ (b) $(2, \infty)$
(c) $(-1, -4/5)$ (d) $(-\infty, -1)$

7. The number of quadratic equations which are unchanged by squaring their roots

- (a) 2 (b) 3
(c) 4 (d) None of these

8. If $b^2 < 2ac$, then equation $ax^3 + bx^2 + cx + d = 0$ has

- (a) exactly one real roots (b) Has three real roots
(c) at least two roots (d) None of these

9. Let α and β are the roots of the equation $x^2 + x + 1 = 0$ then

- (a) $\alpha^2 + \beta^2 = 4$ (b) $(\alpha - \beta)^2 = 3$
(c) $\alpha^3 + \beta^3 = 2$ (d) $\alpha^4 + \beta^4 = 1$

10. If $\sec\alpha, \tan\alpha$ are roots of $ax^2 + bx + c = 0$, then

- (a) $a^4 - b^4 + 4ab^2c = 0$ (b) $a^4 + b^4 - 4ab^2c = 0$
(c) $a^2 - b^2 = 4ac$ (d) $a^2 + b^2 = ac$

11. Let α, β be the roots of $ax^2 + bx + c = 0$, γ, δ be the roots of $px^2 + qx + r = 0$ and D_1 and D_2 be their respective discriminant. If $\alpha, \beta, \gamma, \delta$, are in A.P., then the ratio $D_1 : D_2$ is equal to

- (a) $\frac{a^2}{b^2}$ (b) $\frac{a^2}{p^2}$ (c) $\frac{b^2}{q^2}$ (d) $\frac{c^2}{r^2}$

12. If one root of the equation $(l-m)x^2 + lx + 1 = 0$ is double the other and if l is real, then the greatest value of m is

- (a) $\frac{9}{8}$ (b) $\frac{7}{8}$ (c) $\frac{8}{9}$ (d) $\frac{8}{9}$



13. If p, q, r are real numbers satisfying the condition $p + q + r = 0$, then the roots of the quadratic equation $3px^2 + 5qx + 7r = 0$ are
 (a) Positive (b) Negative
 (c) Real and distinct (d) Imaginary
14. If a, b, c are in G.P., then the equation $ax^2 + 2bx + c = 0$ and $dx^2 + 2ex + f = 0$ have common root if $\frac{d}{a}, \frac{e}{b}, \frac{f}{c}$ are in
 (a) A.P. (b) G.P. (c) H.P. (d) $ab = cd$
15. Sum of values of x and y satisfying the equation $3^x - 4^y = 77$; $3^{x/2} - 2^y = 7$ is:
 (a) 2 (b) 3 (c) 4 (d) 5
16. If the value of $m^4 + \frac{1}{m^4} = 119$, then the value of $\left| m^3 - \frac{1}{m^3} \right| =$
 (a) 11 (b) 18 (c) 24 (d) 36
17. The number of integral roots of the equation $x^8 - 24x^7 - 18x^5 + 39x^2 + 1155 = 0$ is:
 (a) 0 (b) 2 (c) 4 (d) 6
18. If the roots of the equation $\frac{1}{x+p} + \frac{1}{x+q} = \frac{1}{r}$ are equal in magnitude but opposite in sign, then the product of the roots is:
 (a) $-2(p^2 + q^2)$ (b) $-(p^2 + q^2)$
 (c) $-\frac{(p^2 + q^2)}{2}$ (d) $-pq$
19. Minimum possible number of positive root of the quadratic equation $x^2 - (1 + \lambda)x + \lambda - 2 = 0$, $\lambda \in \mathbb{R}$:
 (a) 2 (b) 0
 (c) 1 (d) Can not be determined
20. If $\alpha, \beta, \gamma, \delta \in \mathbb{R}$ satisfy

$$\frac{(\alpha+1)^2 + (\beta+1)^2 + (\gamma+1)^2 + (\delta+1)^2}{\alpha + \beta + \gamma + \delta} = 4$$
 If biquadratic equation $a_0x^4 + a_1x^3 + a_2x^2 + a_3x + a_4 = 0$ has the roots $\left(\alpha + \frac{1}{\beta} - 1\right), \left(\beta + \frac{1}{\gamma} - 1\right), \left(\gamma + \frac{1}{\delta} - 1\right), \left(\delta + \frac{1}{\alpha} - 1\right)$. Then the value of a_2/a_0 is:
 (a) 4 (b) -4
 (c) 6 (d) None the these
21. Both the roots of the equation $(x-b)(x-c) + (x-a)(x-c) + (x-a)(x-b) = 0$ are always
 (a) positive (b) real
 (c) negative (d) None of these.
22. The number of real solutions of the equation $|x|^2 - 3|x| + 2 = 0$ is
 (a) 4 (b) 1 (c) 3 (d) 2
23. Two towns A and B are 60km apart. A school is to be built to serve 150 students in town A and 50 students in town B . If the total distance to be travelled by all 200 student is to be as small as possible, then the school should be built at
 (a) town B (b) 45 km from town A
 (c) town A (d) 45 km from town B
24. If α and β are the roots of $x^2 + px + q = 0$ and α^4, β^4 are the roots of $x^2 - rx + s = 0$, then equation $x^2 - 4qx + 2q^2 - r = 0$ has always
 (a) two real roots
 (b) two positive roots
 (c) two negative roots
 (d) one positive and one negative roots
25. Let a, b, c be real numbers, $a = 0$, if a is roots of $a^2x^2 + bx + c = 0$. β is the root of $a^2x^2 - bx - c = 0$ and $0 < \alpha < \beta$, then the equation $a^2x^2 + 2bx + 2c = 0$ has a root γ that always satisfies
 (a) $\gamma = \frac{\alpha + \beta}{2}$ (b) $\gamma = \alpha + \frac{\beta}{2}$
 (c) $\gamma = \alpha$ (d) $\alpha < \gamma < \beta$
26. The number of points of intersection of two curves $y = 2 \sin x$ and $y = 5x^2 + 2x + 3$ is
 (a) 0 (b) b, c
 (c) 2 (d) $a + c, b + c$
27. If p, q, r are +ve and are in A.P., the roots of quadratic equation $px^2 + qx + r = 0$ are all real for
 (a) $\left| \frac{r}{p} - 7 \right| \geq 4\sqrt{3}$ (b) $\left| \frac{p}{r} - 7 \right| \geq 4\sqrt{3}$
 (c) all p and r (d) no p and r
28. Let $p, q \in \{1, 2, 3, 4\}$. The number of equations of the form $px^2 + qx + 1 = 0$ having real roots is
 (a) 15 (b) 9 (c) 7 (d) 8
29. If α and β ($\alpha < \beta$) are the roots of the equation $x^2 + bx + c = 0$, where $c < 0 < b$, then
 (a) $0 < \alpha < \beta$ (b) $\alpha < 0 < \beta < |\alpha|$
 (c) $\alpha < \beta < 0$ (d) $\alpha < 0 < |\alpha| < \beta$
30. If a, b, c, d are positive real numbers such that $a + b + c + d = 2$, then $M = (a + b)(c + d)$ satisfies the relation
 (a) $0 \leq M \leq 1$ (b) $1 \leq M \leq 2$
 (c) $2 \leq M \leq 3$ (d) $3 \leq M \leq 4$
31. For the equations $3x^2 + px + 3 = 0$, $p > 0$, if one of the roots is square of the other, then p is equal to
 (a) $1/3$ (b) 1 (c) 3 (d) $2/3$
32. If $a_1, a_2, \dots, a_n$ are positive real numbers whose product is a fixed number c , then the minimum value of $a_1 + a_2 + \dots + a_{n-1} + 2a_n$ is
 (a) $n(2c)^{1/n}$ (b) $(n+1)c^{1/n}$
 (c) $2nc^{1/n}$ (d) $(n+1)(2c)^{1/n}$
33. If $\alpha \in \left(0, \frac{\pi}{2}\right)$ then $\sqrt{x^2 + x} + \frac{\tan^2 \alpha}{\sqrt{x^2 + x}}$ is always greater than or equal to
 (a) $2 \tan \alpha$ (b) 1 (c) 2 (d) $\sec^2 \alpha$

34. Number of positive integers n for which $n^2 + 96$ is a perfect square is
 (a) 4 (b) 8 (c) 12 (d) infinite
35. If the equation $ax^2 + 2bx + c = 0$ and $ax^2 + 2cx + b = 0$, $a \neq 0$, $b \neq c$, have a common root then their other roots are the roots of the quadratic equation
 (a) $a^2x(x+1) + 4bc = 0$ (b) $a^2x(x+1) + 8bc = 0$
 (c) $a^2x(x+2) + 8bc = 0$ (d) $a^2x(1+2x) + 8bc = 0$
36. If α, β, γ are the roots of the equation $x^3 + ax + b = 0$ then value of $\frac{\alpha^3 + \beta^3 + \gamma^3}{\alpha^2 + \beta^2 + \gamma^2}$ is equal to:
 (a) $\frac{3b}{2a}$ (b) $\frac{-3b}{2a}$ (c) $3b$ (d) $2b$
37. If $x^2 + 5 = 2x - 4\cos(a + bx)$, where $a, b \in (0, 5)$, is satisfied for at least one real x , then the maximum value of $a + b$ is equal to
 (a) 3π (b) 2π (c) π (d) $-\pi$
38. Let r, s, t be the roots of the equation $x^3 + ax^2 + bx + c = 0$, such that $(rs)^2 + (st)^2 + (rt)^2 = b^2 - kac$, then $k =$
 (a) 1 (b) 2 (c) 3 (d) 4
39. The complete set of values of 'a' for which the inequality $(a-1)x^2 - (a+1)x + (a-1) \geq 0$
 (a) $\left[\frac{3}{7}, 1\right]$ (b) $(-\infty, 1)$ (c) $\left(-\infty, \frac{7}{3}\right]$ (d) $\left[\frac{7}{3}, \infty\right)$
40. The number of integral values of k for which the inequality $x^2 - 2(4k-1)x + 15k^2 - 2k - 7 \geq 0$ hold for all $x \in R$ is:
 (a) 2 (b) 1 (c) 4 (d) infinity
41. The complete set of values of a for which the roots of the equation $x^2 - 2|a+1|x + 1 = 0$ are real is given by:
 (a) $(-\infty, -2] \cup [0, \infty)$ (b) $(-\infty, -1] \cup [0, \infty)$
 (c) $(-\infty, -1] \cup [1, \infty)$ (d) $(-\infty, -2] \cup [1, \infty)$
42. For the quadratic equation $x^2 - (2 + \lambda)x + \lambda - 2 = 0$, $\lambda \in R$
 (a) Minimum possible number of positive root of the equation is 2
 (b) Minimum possible number of positive root of the equation is 1
 (c) Minimum possible number of negative roots of the equation is 1
 (d) Minimum possible number of negative roots of the equation is 2
43. If α, β are the roots of $3x^2 + 2x + 1 = 0$, then the quadratic equation whose en of the root is $\left(\alpha + \frac{1}{\beta}\right)$
 (a) $3x^2 + 8x + 16 = 0$ (b) $3x^2 - 8x + 16 = 0$
 (c) $3x^2 - 8x - 16 = 0$ (d) $3x^2 + 8x - 16 = 0$
44. Let a, b, c, d be distinct real numbers such that a, b are roots or $x^2 - 5cx - 6d = 0$, and c, d are roots of $x^2 - 5ax - 6b = 0$. Then $b + d$ is
 (a) 180 (b) 162 (c) 144 (d) 126
45. Let $S = \{1, 2, 3, \dots, 100\}$. Suppose b and c are chosen at random from the set S . The probability that $4x^2 + bx + c$ has equal roots is
 (a) 0.001 (b) 0.004 (c) 0.007 (d) 0.01
46. Consider the quadratic equation $nx^2 + 7\sqrt{nx} + n = 0$, where n is a positive integer. Which of the following statements are necessarily correct?
 I. For any n , the roots are distinct.
 II. There are infinite many values of n for which both roots are real.
 III. The product of the roots is necessarily an integer.
 (a) III only (b) I and III only
 (c) II and III only (d) I, II and III
47. Let a_1, a_2, a_3, a_4 be real numbers such that $a_1 + a_2 + a_3 + a_4 = 0$ and $a_1^2 + a_2^2 + a_3^2 + a_4^2 = 1$. Then the smallest possible value of the expression $(a_1 - a_2)^2 + (a_2 - a_3)^2 + (a_3 - a_4)^2 + (a_4 - a_1)^2$ lies in the interval
 (a) (0, 1.5) (b) (1.5, 2.5) (c) (2.5, 3) (d) (3, 3.5)
48. Let $r(x)$ be the remainder when the polynomial $x^{135} + x^{125} - x^{115} + x^5 + 1$ is divided by $x^3 - x$. Then
 (a) $r(x)$ is the zero polynomial
 (b) $r(x)$ is a non-zero constant
 (c) degree of $r(x)$ is one
 (d) degree of $r(x)$ is two
49. The sum of all non-integer roots of the equation $x^5 - 6x^4 + 11x^3 - 5x^2 - 3x + 2 = 0$ is
 (a) 6 (b) -11 (c) -5 (d) 3
50. If $a + b + c = 5$, $a^2 + b^2 + c^2 = 12$ and $a^3 + b^3 + c^3 = 25$. Then the value of $a^4 + b^4 + c^4$ is
 (a) $\frac{251}{6}$ (b) $\frac{253}{6}$
 (c) $\frac{255}{6}$ (d) $\frac{257}{6}$
51. The least value of the expression $x^2 + 4y^2 + 3z^2 - 2x - 12y - 6z + 14$ is
 (a) 1 (b) no least value
 (c) 0 (d) None of these
52. If $P(x) = ax^2 + bx + c$ leaves a remainder of 4 when divided by x , a remainder of 3 when divided by $(x+1)$ and a remainder of 1 when divided by $(x-1)$ then $P(2)$ is
 (a) 3 (b) 6 (c) -3 (d) -6
53. If the equations $px^2 + qx + r = 0$ and $rx^2 + qx + p = 0$ ($p \neq r \neq 0$) have a negative common root then the value of $(p - q + r)$ is equal to
 (a) -1 (b) 1 (c) 2 (d) 0

54. Let α and β two real roots of the equation $(k+1)\tan^2 x - \sqrt{2} \cdot \lambda \tan x = (1-k)$, where $k (\neq -1)$ and λ are real numbers. If $\tan^2(\alpha + \beta) = 50$, then a value of λ is
 (a) 5 (b) 10 (c) $10\sqrt{2}$ (d) $5\sqrt{2}$
55. Let α and β be the roots of the equation, $5x^2 + 6x - 2 = 0$. If $S_n = \alpha^n + \beta^n$, $n = 1, 2, 3, \dots$, then
 (a) $5S_6 + 6S_5 + 2S_4 = 0$ (b) $6S_6 + 5S_5 = 2S_4$
 (c) $6S_6 + 5S_5 + 2S_4 = 0$ (d) $5S_6 + 6S_5 = 2S_4$
56. Let $\lambda \neq 0$ be in R . If α and β are the roots of the equation, $x^2 - x + 2\lambda = 0$ and α and γ are the roots of the equation, $3x^2 - 10x + 27\lambda = 0$, then $\frac{\beta\gamma}{\lambda}$ is equal to
 (a) 27 (b) 9 (c) 18 (d) 36
57. Let p and q be two positive numbers such that $p + q = 2$ and $p^4 + q^4 = 272$. Then p and q are roots of the equation
 (a) $x^2 - 2x + 2 = 0$ (b) $x^2 - 2x + 8 = 0$
 (c) $x^2 - 2x + 136 = 0$ (d) $x^2 - 2x + 16 = 0$
58. Let α and β be the roots of $x^2 - 6x - 2 = 0$. If $a_n = \alpha^n - \beta^n$ for $n \geq 1$, then the value of $\frac{a_{10} - 2a_8}{3a_9}$
 (a) 2 (b) 1 (c) 4 (d) 3
59. If $\alpha, \beta, \gamma, \delta \in R$ satisfy $\frac{(\alpha-1)^2 + (\beta+1)^2 + (\gamma+1)^2 + (\delta+1)^2}{\alpha + \beta + \gamma + \delta} = 4$
 if biquadratic equation $a_0x^4 + a_1x^3 + a_2x^2 + a_3x + a_4 = 0$ has the roots $\left(\alpha + \frac{1}{\beta} - 1\right), \left(\beta + \frac{1}{\gamma} - 1\right), \left(\gamma + \frac{1}{\delta} - 1\right), \left(\delta + \frac{1}{\alpha} - 1\right)$.
 Then the value of $\frac{a_2}{a_0}$ is
 (a) 4 (b) -4
 (c) 6 (d) None of these
60. The number of all 3-digit integers abc for which $(a \times b \times c) + (a \times b)6 + (c \times a) + a + b + c = 29$ is
 (a) 6 (b) 10 (c) 14 (d) 18

Subjective Type Question

61. Solve for x : $x^{\frac{1}{3}} + (2x-3)^{\frac{1}{3}} = \{12(x-1)\}^{\frac{1}{3}}$.
62. Find two positive numbers such that their sum multiplied by the sum of their squares is 5500, and their difference multiplied by the difference of their squares is 352.
63. Solve: $\sqrt[3]{6(5x+6)} - \sqrt[3]{5(6x-11)} = 1$.
64. Find two numbers such that their sum is 9, and the sum of their fourth powers 2417.
65. Solve
 (i) $\sqrt{4x^2 - 7x - 15} - \sqrt{x^2 - 3x} = \sqrt{x^2 - 9}$.
 (ii) $\sqrt{2x^2 - 9x - 4} + 3\sqrt{2x-1} = \sqrt{2x^2 + 21x - 11}$.

$$(iii) \sqrt{2x^2 + 5x - 7} + \sqrt{3(x^2 - 7x + 6)} - \sqrt{7x^2 - 6x - 1} = 0.$$

$$(iv) \sqrt{a^2 + 2ax - 3x^2} - \sqrt{a^2 + ax - 6x^2} = \sqrt{2a^2 + 3ax - 9x^2}.$$

66. Let p and q be the two roots of the equation $mx^2 + x(2-m) + 3 = 0$. If m_1, m_2 are the two values of m satisfying $\frac{p}{q} + \frac{q}{p} = \frac{2}{3}$ the value of $\frac{m_1}{m_2} + \frac{m_2}{m_1}$ is
67. If $c > 0$ and $2ax^2 + 3bx + 5c = 0$ does not have any real roots, then prove that $2a - 3b + 5c > 0$.
68. If $ax^2 + bx + 6 = 0$ does not have distinct real roots, then find the least value of $3a + b$.
69. The minimum number of real roots of $(x^2 + 3x + a)(x^2 + ax + 1) = 0$ equals
70. The value of k for which $kx^2 - 2kx + 3x - 6$ is positive for exactly two integral values of x is
71. The number of order pairs of integers (x, y) satisfying the equation $x^2 + 6x + y^2 = 4$ is
72. Suppose that $2 - \sqrt{95}$ is a root of $x^2 + ax + b = 0$ where b is negative real number and a is a integer. The largest possible value of a , is
73. Given that α, β, γ are the roots of cubic equation $x^3 = 3x^2 + 2x + \frac{2}{3} = 0$. The value of $\alpha^4 + \beta^4 + \gamma^4$ is equal to _____
74. Let $P(x)$ be a polynomial whose coefficient are integer satisfies $P(1) = 5$ and $P(2) = 7$. The smallest possible positive value of $P(10)$ is equal to _____
75. Let α and β be two real numbers such that $\alpha + \beta = 1$ and $\alpha\beta = -1$. Let $p_n = (\alpha)^n + (\beta)^n$, $p_{n-1} = 11$ and $p_{n+1} = 29$ for some integer $n \geq 1$. Then, the value of p_n^2 is _____.
76. Let the quadratic equation $x^2 + 3x - k = 0$ has roots a, b and $x^2 + 3x - 10 = 0$ has roots c, d such that modulus of difference of the roots of the first equation is equal to twice the modulus of the difference of the roots of the second equation. If the value of ' k ' can be expressed as rational number in the lowest form as m/n then find the value of $(m+n)$.
77. When $y^2 + my + 2$ is divided by $(y-1)$ then the quotient is $f(y)$ and the remainder is R_1 . When $y^2 + my + 2$ is divided by $(y+1)$ then quotient is $g(y)$ and the remainder is R_2 . If $R_1 = R_2$ then find the value of m .
78. Let $f(x)$ be the remainder when $p(x) = x^{73} - 2x^6 + 3x^3$ is divided by $x^4(x^2 - 1)$ then $f(1) =$
79. Find all the values of ' a ', so that exactly one root of the equation $x^2 - 2ax + a^2 - 1 = 0$, lies between the numbers 2 and 4, and no root of the equation is either equal to 2 or equal to 4.

Exercise-3 (JEE Advanced Level)

MULTIPLE CORRECT TYPE QUESTIONS

1. Let α, β be the roots of the equation $x^2 - px + r = 0$ and $\frac{\alpha}{2}, 2\beta$ be the roots of the equation $x^2 - qx + r = 0$. Then the value of r is

(a) $\frac{2}{9}(p-q)(2q-p)$ (b) $\frac{2}{9}(q-p)(2p-q)$
 (c) $\frac{2}{9}(q-2p)(2q-q)$ (d) $\frac{2}{9}(2p-q)(2q-q)$

2. Let p and q be real numbers such that $p = 0, p^3 = q$ and $p^3 = -q$. If α and β are nonzero complex numbers satisfying $\alpha + \beta = -p$ and $\alpha^3 + \beta^3 = q$, then a quadratic equation having $\frac{\alpha}{\beta}$ and $\frac{\beta}{\alpha}$ as its roots is

(a) $(p^3 + q)x^2 - (p^3 + 2q)x + (p^3 + q) = 0$
 (b) $(p^3 + q)x^2 - (p^3 - 2q)x + (p^3 + q) = 0$
 (c) $(p^3 - q)x^2 - (5p^3 - 2q)x + (p^3 - q) = 0$
 (d) $(p^3 - q)x^2 - (5p^3 + 2q)x + (p^3 - q) = 0$

3. A value of b for which the equations

$$x^2 + bx - 1 = 0$$

$$x^2 + x + b = 0$$

have one root in common is

(a) $-\sqrt{2}$ (b) $-i\sqrt{3}$
 (c) $i\sqrt{5}$ (d) $\sqrt{2}$

4. Values of k for which the inequality $k\sin^2 x - k\sin x + 1 \geq 0$ is true $\forall x \in R$ is

(a) $k > -\frac{1}{2}$ (b) $k > 4$
 (c) $-\frac{1}{2} \leq k \leq 4$ (d) $\frac{1}{2} \leq k \leq 5$

5. Suppose the quadratic polynomial $P(x) = ax^2 + bx + c$ has positive coefficients a, b, c in arithmetic progression in that order. If $P(x) = 0$ has integer roots α and β , then $\alpha + \beta + \alpha\beta$ equals

(a) 3 (b) 5 (c) 7 (d) 14

6. $f(x) = \frac{x^4 - 14x^2 + 25}{x - \frac{5}{x} + 2}$, zeroes are in the form $a \pm \sqrt{b}$,

$a, b \in z$ then $a + b$ is equal to

(a) 5 (b) 6
 (c) 7 (d) 8

7. Let $\alpha = \max_{x \in R} \{8^{2\sin 3x} \cdot 4^{4\cos 3x}\}$ and $\min_{x \in R} \{8^{2\sin 3x} \cdot 4^{4\cos 3x}\}$.

If $8x^2 + bx + c = 0$ is a quadratic equation whose roots are $\alpha^{1/5}$ and $\beta^{1/5}$, then the value of $c - b$ is equal to

(a) 42 (b) 43 (c) 47 (d) 50

8. If $(x^2 + x + 1) + (x^2 + 2x + 3) + (x^2 + 3x + 5) + \dots + (x^2 + 20x + 39) = 4500$, then x is equal to

(a) 10 (b) -10 (c) 20.5 (d) -20.5

9. If the equation whose roots are the squares of the roots of the cubic $x^3 - ax^2 + bx - 1 = 0$ is identical with the given cubic equation, then

(a) $a = 0, b = 3$
 (b) $a = b = 0$
 (c) $a = b = 3$
 (d) a, b are roots of $x^2 + x + 2 = 0$

10. If the quadratic equation $ax^2 + bx + c = 0$ ($a > 0$) has $\sec^2 \theta$ and $\csc^2 \theta$ as its roots, then which of the following must hold good?

(a) $b + c = 0$ (b) $b^2 - 4ac \geq 0$
 (c) $c \geq 4a$ (d) $4a + b \geq 0$

11. If $x \in R$, then the roots of the equation $x^4 + 4x^3 - 8x^2 + k = 0$ with respect to the values of k , ($k \in [0, 2014]$), are

(a) for $k \in [0, 3]$: 4 real and distinct roots
 (b) for $k \in (3, 128)$: two real and two imaginary roots
 (c) for $k \in (128, 583)$: no real root
 (d) for $k \in (0, 3)$: four real and distinct roots

12. For the quadratic polynomial $f(x) = 4x^2 - 8kx + k$, the statements which hold good are

(a) there is only one integral k for which $f(x)$ is non-negative $\forall x \in R$
 (b) for $k < 0$ the number zero lies between the zeros of the polynomial
 (c) $f(x) = 0$ has two distinct solutions in $(0, 1)$ for $k \in (1/4, 4/7)$
 (d) Minimum value of $f(x) \forall k \in R$ is $k(1 + 12k)$

13. Let α and β be the roots of the equation $x^2 - x - 1 = 0$. If $p_k = (\alpha)^k + (\beta)^k$, $k \geq 1$, then which one of the following statements is not true?

(a) $p_5 = p_2 \cdot p_3$
 (b) $p_3 = p_5 - p_4$
 (c) $(p_1 + p_2 + p_3 + p_4 + p_5) = 26$
 (d) $p_5 = 11$

COMPREHENSION BASED QUESTIONS

Comprehension (Q. No. 14 to 16)

Consider the equation $x^4 - \lambda x^2 + 9 = 0$. This can be solved by substituting $x^2 = t$ such equations are called as pseudo quadratic equations.

14. If the equation has four real and distinct roots, then λ lies in the interval

(a) $(-\infty, -6) \cup (6, \infty)$ (b) $(0, \infty)$
(c) $(6, \infty)$ (d) $(-\infty, -6)$

15. If the equation has no real root, then λ lies in the interval

(a) $(-\infty, 0)$ (b) $(-\infty, 6)$
(c) $(6, \infty)$ (d) $(0, \infty)$

16. If the equation has only two real roots, then set of values of λ is

(a) $(-\infty, -6)$ (b) $(-6, 6)$
(c) $\{6\}$ (d) ϕ

Comprehension (Q. No. 17 to 19) Consider the equation $x^4 - \lambda x^2 + 9 = 0$.

17. If the equation has 4 real and distinct roots, then λ lies in the interval.

(a) $(-\infty, -6) \cup (6, \infty)$ (b) $(0, \infty)$
(c) $(6, \infty)$ (d) $(-\infty, -6)$

18. If the equation has no real root, then λ lies in the interval

(a) $(-\infty, 0)$ (b) $(-\infty, 6)$
(c) $(6, \infty)$ (d) $(0, \infty)$

19. If the equation has only 2 real roots, then the set of values of λ is

(a) $(-\infty, -6)$ (b) $(-6, 6)$
(c) $\{6\}$ (d) ϕ

Comprehension (Q. No. 20 to 22) Consider the equation $x^4 + 2ax^3 + x^2 + 2ax + 1 = 0$, where $a \in \mathbb{R}$. Also, range of $f(x) = x + \frac{1}{x}$ is $(-\infty, -2] \cup [2, \infty)$

20. If equation has at least two distinct positive real roots, then all possible value of 'a' are

(a) $\left(-\infty, -\frac{1}{4}\right)$ (b) $\left(\frac{5}{4}, \infty\right)$
(c) $\left(-\infty, -\frac{3}{4}\right)$ (d) $\left(-\frac{5}{4}, \infty\right)$

21. If equation has at least two distinct negative real roots, then all possible values of 'a' are

(a) $\left(\frac{3}{4}, \infty\right)$ (b) $\left(-\frac{5}{4}, \infty\right)$
(c) $\left(-\infty, \frac{1}{4}\right)$ (d) All of above

22. If exactly two roots are positive and two roots are negative, then number of integral values of 'a' is

(a) 2 (b) 1
(c) 0 (d) 3

MATCH THE COLUMN TYPE QUESTIONS

23. Match the column.

Column-I		Column-II	
A.	If a, b, c , and d are four zero real number such that $(d + a - b)^2 + (d + b - c)^2 = 0$ and the roots of the equation $a(b - c)x^2 + b(c - a)x + c(a - b) = 0$ are real and equal. Then	p.	$a + b + c = 0$
B.	If the roots of the equation $(a^2 + b^2)x^2 - 2b(a + c)x + (b^2 + c^2) = 0$ are real and equal, then	q.	a, b, c are in A.P
C.	If the equation $ax^2 + bx + c = 0$ and $x^3 - 3x^2 + 3x - 1 = 0$ have a common real root, then	r.	a, b, c are in G.P
D.	Let a, b, c be positive real number such that the expression $bx^2 + (\sqrt{(a+c)^2 + 4b^2})x + (a+c)$ is non-negative $\forall x \in \mathbb{R}$, then	s.	a, b, c are in H.P

- (a) $A \rightarrow s, B \rightarrow r, C \rightarrow p, D \rightarrow q$
(b) $A \rightarrow q, s, B \rightarrow p, C \rightarrow r, D \rightarrow p$
(c) $A \rightarrow q, s, B \rightarrow r, C \rightarrow p, D \rightarrow p$
(d) $A \rightarrow p, B \rightarrow r, C \rightarrow p, D \rightarrow q, s$

24. Match the column.

Column-I		Column-II	
A.	one roots is positive and the other is negative for the equation $(m-2)x^2 - (8-2m)x - (8-3m) = 0$, then number of Integral values of m	p.	0
B.	exactly one root of equation $x^2 - m(2x - 8) - 15 = 0$ lies in interval $(0, 1)$	q.	infinity solution
C.	the equation $x^2 + 2(m+1)x + 9m - 5 = 0$ has both roots negative	r.	1
D.	the equation $x^2 + 2(m-1)x + m + 5 = 0$ has both roots lying on either sides of 1	s.	2

- (a) $A \rightarrow q, B \rightarrow r, C \rightarrow q, D \rightarrow p$
(b) $A \rightarrow p, B \rightarrow r, C \rightarrow q, D \rightarrow q$
(c) $A \rightarrow p, B \rightarrow r, C \rightarrow q, D \rightarrow s$
(d) $A \rightarrow p, B \rightarrow q, C \rightarrow r, D \rightarrow q$

NUMERICAL BASED QUESTIONS

25. If a, b are the roots of $x^2 + px + 1 = 0$ and c, d are the roots of $x^2 + qx + 1 = 0$. Then find the value of $(a - c)(b - c)(a + d)(b + d)/(q^2 - p^2)$.
26. Let α, β be the roots of the equation $x^2 + ax + b = 0$ and γ, δ be the roots of $x^2 - ax + b - 2 = 0$. If $\alpha\beta\gamma\delta = 24$ and $\frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma} + \frac{1}{\delta} = \frac{5}{6}$, then find the value of a .
27. If $a > b > 0$ and $a^3 + b^3 + 27ab = 729$ then the quadratic equation $ax^2 + bx - 9 = 0$ has roots α, β ($\alpha < \beta$). Find the value of $4\beta - \alpha$.
28. All the value of k for which the quadratic polynomial $f(x) = -2x^3 + kx + k^2 + 5$ has two distinct zeroes and or one of them satisfying $0 < x < 2$, lie in the interval (a, b) . The value of $(a + 10b)$ is _____.
29. The quadratic equation $x^2 + mx + n = 0$ has roots which are twice those of $x^3 + px + m = 0$ and m, n and $p = q$. Then the value of n/p is _____.
30. Let a, b and c be real numbers which satisfy the equation $a + \frac{1}{bc} = \frac{1}{5}$, $b + \frac{1}{ac} = \frac{-1}{15}$, and $c + \frac{1}{ab} = \frac{1}{ab}$. The value of $\frac{c-b}{c-a}$ is equal to _____.
31. If a, b, c are non-zero real number, then the minimum value of the expression
$$\left(\frac{(a^4 + 3a^2 + 1)(b^4 + 5b^2 + 1)(c^4 + 7c^2 + 1)}{a^2 b^2 c^2} \right)$$
32. If $a, b \in \mathbb{R}$ such that $a + b = 1$ and $(1 - 2ab)(a^3 + b^3) = 12$. The value of $(a^2 + b^2)$ is equal _____.
33. If the cubic $2x^3 - 9x^2 + 12x + k = 0$ has two equal roots then maximum value of $|k|$ is _____.
34. Let a, b and c be distinct non-zero real numbers such that $\frac{1-a^3}{a} = \frac{1-b^3}{b} = \frac{1-c^3}{c}$. The value of $(a^3 + b^3 + c^3)$ is _____.
35. If the fraction $\frac{x^3 + (a-10)x^2 - x + a - 6}{x^3 + (a-6)x^2 - x + a - 10}$ reduces to quotient of two linear functions, then a equals _____.
36. If $a + b + c = 0$, $a^2 + b^2 + c^2 = 4$, then $a^4 + b^4 + c^4$.
37. Let $f(x)$ be the remainder when $p(x) = x^{73} - 2x^6 + 3x^3$ is divided by $x^4(x^2 - 1)$ then find the value of $f(1)$
38. If one root of the equation $ax^2 + bc + c = 0$ is equal to n^{th} power of the other root, then show that $(ac^n)^{1/n(n+1)} + (a^n c)^{1/(n+1)} + b = 0$
39. Let p & q are real numbers such that $(\tan p - 1)^3 + 2019(\tan p - 1) = 1$ & $(1 - \cot q)^3 + 2019(1 - \cot q) = -1$ $\tan p \neq \cot q$, then number of possible values of r which satisfy the equation $\tan p + \cot q + \sin r + \cos r = 3$ in $x \in [-2\pi, 2\pi]$ is _____.
40. If the quadratic equation $x^2 + (2 - \tan \theta)x - (1 + \tan \theta) = 0$ has two integral roots, then sum of all possible values of θ in the interval $(0, 2\pi)$ is $k\pi$, then the value of k is equal to _____.

Exercise-4 (Past Year Questions)

JEE MAIN

1. The sum of all real values of x satisfying the equation $(x^2 - 5x + 5)^{x^2 + 4x - 60} = 1$ is: (2016)
 (a) 3 (b) -4
 (c) 6 (d) 5
2. If, for a positive integer n , the quadratic equation, $x(x + 1) + (x + 1)(x + 2) + \dots + (x + n - 1)(x + n) = 10n$ has two consecutive integral solutions, then n is equal to: (2017)
 (a) 11 (b) 12 (c) 9 (d) 10
3. If $\alpha, \beta \in \mathbb{C}$ are the distinct roots of the equation $x^2 - x + 1 = 0$, then $\alpha^{101} + \beta^{107}$ is equal to (2018)
 (a) 0 (b) 1
 (c) 2 (d) -1

4. Let α and β be two roots of the equation $x^2 + 2x + 2 = 0$, then $\alpha^{15} + \beta^{15}$ is equal to: (2019)
 (a) -256 (b) 512
 (c) -512 (d) 256
5. If both the roots of the quadratic equation $x^2 - mx + 4 = 0$ are real and distinct and they lie in the interval $[1, 5]$, then m lies in the interval (2019)
 (a) (4, 5) (b) (3, 4)
 (c) (5, 6) (d) (-5, -4)
6. The number of all possible positive integral values of α for which the roots of the quadratic equation, $6x^2 - 11x + \alpha = 0$ are rational numbers is: (2019)
 (a) 2 (b) 5
 (c) 3 (d) 4
7. Consider the quadratic equation $(c - 5)x^2 - 2cx + (c - 4) = 0$, $c \neq 5$. Let S be the set of all integral values of c for which

one root of the equation lies in the interval $(0, 2)$ and its other root lies in the interval $(2, 3)$. Then the number of elements in S is: **(2019)**

- (a) 18 (b) 12
(c) 10 (d) 11

8. If one real root of the quadratic equation $81x^2 + kx + 256 = 0$ is cube of the other root, then a value of k is: **(2019)**

- (a) -81 (b) 100
(c) 144 (d) -300

9. The number of integral values of m for which the quadratic expression, $(1 + 2m)x^2 - 2(1 + 3m)x + 4(1 + m)$, $x \in \mathbb{R}$, is always positive, is: **(2019)**

- (a) 3 (b) 8
(c) 7 (d) 6

10. If λ be the ratio of the roots of the quadratic equation in x , $3m^2x^2 + m(m - 4)x + 2 = 0$, then the least value of m for which $\lambda + \frac{1}{\lambda} = 1$, is **(2019)**

- (a) $2 - \sqrt{3}$ (b) $4 - 3\sqrt{2}$
(c) $-2 + \sqrt{2}$ (d) $4 - 2\sqrt{3}$

11. If α and β be the roots of the equation $x^2 - 2x + 2 = 0$, then the least value of n for which $\left(\frac{\alpha}{\beta}\right)^n = 1$ is **(2019)**

- (a) 2 (b) 3
(c) 4 (d) 5

12. If three distinct numbers a, b, c are in G.P. and the equations $ax^2 + 2bx + c = 0$ and $dx^2 + 2ex + f = 0$ have a common root, then which one of the following statements is correct? **(2019)**

- (a) d, e, f are in A.P. (b) $\frac{d}{a}, \frac{e}{b}, \frac{f}{c}$ are in G.P.
(c) $\frac{d}{a}, \frac{e}{b}, \frac{f}{c}$ are in A.P. (d) d, e, f are in G.P.

13. The number of integral values of m for which the equation $(1 + m^2)x^2 - 2(1 + 3m)x + (1 + 8m) = 0$ has no real root is: **(2019)**

- (a) infinitely many (b) 2
(c) 3 (d) 1

14. Let $p, q \in \mathbb{R}$. If $2 - \sqrt{3}$ is a root of the quadratic equation, $x^2 + px + q = 0$, then: **(2019)**

- (a) $q^2 + 4p + 14 = 0$ (b) $p^2 - 4q - 12 = 0$
(c) $q^2 - 4p - 16 = 0$ (d) $p^2 - 4q + 12 = 0$

15. If m is chosen in the quadratic equation $(m^2 + 1)x^2 - 3x + (m^2 + 1)^2 = 0$ such that the sum of its roots is greatest, then the absolute difference of the cubes of its roots is: **(2019)**

- (a) $8\sqrt{3}$ (b) $4\sqrt{3}$
(c) $10\sqrt{5}$ (d) $8\sqrt{5}$

16. If α and β are the roots of the quadratic equation, $x^2 + x\sin\theta - 2\sin\theta = 0$, $\theta \in \left(0, \frac{\pi}{2}\right)$, then $\frac{\alpha^{12} + \beta^{12}}{(\alpha^{-12} + \beta^{-12})(\alpha - \beta)^{24}}$ is equal to: **(2019)**

- (a) $\frac{2^6}{(\sin\theta + 8)^{12}}$ (b) $\frac{2^{12}}{(\sin\theta - 8)^6}$
(c) $\frac{2^{12}}{(\sin\theta - 4)^{12}}$ (d) $\frac{2^{12}}{(\sin\theta + 8)^{12}}$

17. The number of real roots of the equation $5 + |2^x - 1| = 2^x$ ($2^x - 2$) is: **(2019)**

- (a) 2 (b) 3
(c) 4 (d) 1

18. If α and β are the roots of the equation $375x^2 - 25x - 2 = 0$, then $\lim_{n \rightarrow \infty} \sum_{r=1}^n \alpha^r + \lim_{n \rightarrow \infty} \sum_{r=1}^n \beta^r$ is equal to: **(2019)**

- (a) $\frac{21}{346}$ (b) $\frac{29}{358}$
(c) $\frac{1}{12}$ (d) $\frac{7}{116}$

19. If α, β and γ are three consecutive terms of a non-constant G.P. such that the equations $\alpha x^2 + 2\beta x + \gamma = 0$ and $x^2 + x - 1 = 0$ have a common root, then $\alpha(\beta + \gamma)$ is equal to: **(2019)**

- (a) $\beta\gamma$ (b) 0
(c) $\alpha\gamma$ (d) $\alpha\beta$

20. Let α and β be the roots of the equation $x^2 - x - 1 = 0$. If $p_k = (\alpha)^k + (\beta)^k$, $k \geq 1$, then which of the following statements is not true? **(2020)**

- (a) $p_5 = p_2 \cdot p_3$ (b) $(p_1 + p_2 + p_3 + p_4 + p_5) = 26$
(c) $p_3 = p_5 - p_4$ (d) $p_5 = 11$

21. The least positive value of 'a' for which the equation, $2x^2 + (a - 10)x + \frac{33}{2} = 2a$ has real roots is _____. **(2020)**

22. Let S be the set of all real roots of the equation, $3^x(3^x - 1) + 2 = |3^x - 1| + |3^x - 2|$. Then S : **(2020)**

- (a) contains exactly two elements
(b) is a singleton
(c) contains at least four elements
(d) is an empty set

23. The number of real roots of the equation, $e^{4x} + e^{3x} - 4e^{2x} + e^x + 1 = 0$ **(2020)**

- (a) 3 (b) 1
(c) 4 (d) 2

24. Let $a, b \in \mathbb{R}$, $a \neq 0$ be such that the equation, $ax^2 - 2bx + 5 = 0$ has a repeated root α , which is also a root of the equation, $x^2 - 2bx - 10 = 0$. If β is the other root of this equation, $\alpha^2 + \beta^2$ is equal to: (2020)
- (a) 25 (b) 26
(c) 24 (d) 28
25. Let $f(x)$ be a quadratic polynomial such that $f(-1) + f(b) = 0$. If one of the roots of $f(x) = 0$ is 3, then its other root lies in: (2020)
- (a) $(-1, 0)$ (b) $(-3, -1)$
(c) $(0, 1)$ (d) $(1, 3)$
26. The set of all real values of λ for which the quadratic equations, $(\lambda^2 + 1)x^2 - 4\lambda x + 2 = 0$ always have exactly one root in the interval $(0, 1)$ is: (2020)
- (a) $(-3, -1)$ (b) $(2, 4]$
(c) $(0, 2)$ (d) $(1, 3]$
27. If α and β are the roots of the equation $x^2 + px + 2 = 0$ and $\frac{1}{\alpha}$ and $\frac{1}{\beta}$ are the roots of the equation $2x^2 + 2qx + 1 = 0$, then $\left(\alpha - \frac{1}{\alpha}\right)\left(\beta - \frac{1}{\beta}\right)\left(\alpha + \frac{1}{\beta}\right)\left(\beta + \frac{1}{\alpha}\right)$ is equal to: (2020)
- (a) $\frac{9}{4}(9 - q^2)$ (b) $\frac{9}{4}(9 + p^2)$
(c) $\frac{9}{4}(9 + q^2)$ (d) $\frac{9}{4}(9 - p^2)$
28. Let α and β be the roots of $x^2 - 3x + p = 0$ and γ and δ be the roots of $x - 6x + q = 0$. If $\alpha, \beta, \gamma, \delta$ form a geometric progression. Then ratio $(2q + p) : (2q - p)$ is: (2020)
- (a) 3 : 1 (b) 5 : 3
(c) 9 : 7 (d) 33 : 31
29. If α and β are the roots of the equation, $7x^2 - 3x - 2 = 0$, then the value of $\frac{\alpha}{1 - \alpha^2} + \frac{\beta}{1 - \beta^2}$ is equal to: (2020)
- (a) $\frac{1}{24}$ (b) $\frac{27}{32}$
(c) $\frac{3}{8}$ (d) $\frac{27}{16}$
30. The product of the roots of the equation $9x^2 - 18|x| + 5 = 0$, is: (2020)
- (a) $\frac{25}{9}$ (b) $\frac{25}{81}$
(c) $\frac{5}{9}$ (d) $\frac{5}{27}$
31. If α and β be two roots of the equation $x^2 - 64x + 256 = 0$. Then the value of $\left(\frac{\alpha^3}{\beta^5}\right)^{\frac{1}{8}} + \left(\frac{\beta^3}{\alpha^5}\right)^{\frac{1}{8}}$ is: (2020)
- (a) 3 (b) 2
(c) 4 (d) 1
32. Let a, b, c be in arithmetic progression. Let the centroid of the triangle with vertices (a, c) , $(2, b)$ and (a, b) be $\left(\frac{10}{3}, \frac{7}{3}\right)$. If α, β are the roots of the equation $ax^2 + bx + 1 = 0$, then the value of $\alpha^2 + \beta^2 - \alpha\beta$ is: (2021)
- (a) $\frac{71}{256}$ (b) $-\frac{69}{256}$
(c) $\frac{69}{256}$ (d) $-\frac{71}{256}$
33. The number of the real roots of the equation $(x + 1)^2 + |x - 5| = \frac{27}{4}$ is (2021)
34. The coefficients a, b and c of the quadratic equation, $ax^2 + bx + c = 0$ are obtained by throwing a dice three times. The probability that this equation has equal roots is (2021)
- (a) $\frac{1}{54}$ (b) $\frac{1}{72}$
(c) $\frac{1}{36}$ (d) $\frac{5}{216}$
35. The integer 'k', for which the inequality $x^2 - 2(3k - 1)x + 8k^2 - 7 > 0$ is valid for every x in $\mathbb{R}$ is: (2021)
- (a) 3 (b) 2
(c) 4 (d) 0
36. If $\alpha, \beta \in \mathbb{R}$ are such that $1 - 2i$ (here $i^2 = -1$) is a root of $z^2 + az + \beta = 0$, then $(\alpha - \beta)$ is equal to: (2021)
- (a) 7 (b) -3
(c) 3 (d) -7
37. Let α and β be the roots of $x^2 - 6x - 2 = 0$. If $a_n = \alpha^n - \beta^n$ for $n \geq 1$, then the value of $\frac{a_{10} - 2a_8}{3a_9}$ is: (2021)
- (a) 4 (b) 1 (c) 2 (d) 3
38. The number of solutions of the equation $\log_4(x - 1) = \log_2(x - 3)$ is (2021)
39. The number of elements in the set $\{x \in \mathbb{R} : (|x| - 3)|x + 4| = 6\}$ is equal to (2021)
- (a) 3 (b) 2 (c) 4 (d) 1

40. Let $P(x) = x^2 + bx + c$ be a quadratic polynomial with real coefficients such that $\int_0^1 P(x) dx = 1$ and $P(x)$ leaves remainder 5 when it is divided by $(x-2)$. Then the value of $9(b+c)$ is equal to: (2021)

(a) 9 (b) 15
(c) 7 (d) 11

41. The value of $4 + \frac{1}{5 + \frac{1}{4 + \frac{1}{5 + \frac{1}{4 + \dots \infty}}}}$ (2021)

(a) $2 + \frac{2}{5}\sqrt{30}$ (b) $2 + \frac{4}{\sqrt{5}}\sqrt{30}$
(c) $4 + \frac{4}{\sqrt{5}}\sqrt{30}$ (d) $5 + \frac{2}{5}\sqrt{30}$

42. If $f(x)$ and $g(x)$ are two polynomials such that the polynomial $P(x) = f(x^3) + xg(x^3)$ is divisible by $x^2 + x + 1$, then $P(1)$ is equal to— (2021)

43. If the sum of the squares of the reciprocals of the roots α and β of the equation $3x^2 + \lambda x - 1 = 0$ is 15, then $6(\alpha^3 + \beta^3)^2$ is equal to (2022)

(a) 18 (b) 24
(c) 36 (d) 96

44. Let $a, b \in \mathbb{R}$ be such that the equation $ax^2 - 2bx + 15 = 0$ has a repeated root α . If α and β are the roots of the equation $x^2 - 2bx + 21 = 0$, then $\alpha^2 + \beta^2$ is equal to: (2022)

(a) 37 (b) 58
(c) 68 (d) 92

45. If the sum of all the roots of the equation $e^{2x} - 11e^x - 45e^{-x} + \frac{81}{2} = 0$ is $\log_e P$, then p is equal to _____. (2022)

46. Let α, β be the roots of the equation $x^2 - 4\lambda x + 5 = 0$ and α, γ be the roots of the equation $x^2 - (3\sqrt{2} + 2\sqrt{3})x + 7 + 3\lambda\sqrt{3} = 0$. If $\beta + \gamma = 3\sqrt{2}$, then $(\alpha + 2\beta + \gamma)^2$ is equal to (2022)

JEE ADVANCED

47. The largest interval for which $x^{12} - x^9 + x^4 - x + 1 > 0$ is (1982)

(a) $-4 < x \leq 0$ (b) $0 < x < 1$
(c) $-100 < x < 100$ (d) $-\infty < x < \infty$

48. The equation $x^4 \frac{3(\log_2 x)^2 + \log_2 x - 5}{4} = \sqrt{2}$ has (1989)

(a) atleast one real solution
(b) exactly three real solutions
(c) exactly one irrational solution
(d) complex roots

49. The quadratic equation $p(x) = 0$ with real coefficients has purely imaginary roots. Then the equation $p(p(x)) = 0$ has (2014)

(a) only purely imaginary roots
(b) all real roots
(c) two real and two purely imaginary roots
(d) neither real nor purely imaginary roots

50. Let S be the set of all non-zero real numbers α such that the quadratic equation $\alpha x^2 - x + \alpha = 0$ has two distinct real roots x_1 and x_2 satisfying the inequality $|x_1 - x_2| < 1$. Which of the following intervals is(are) a subset(s) of S ? (2015)

(a) $\left(-\frac{1}{2}, -\frac{1}{\sqrt{5}}\right)$ (b) $\left(-\frac{1}{\sqrt{5}}, 0\right)$
(c) $\left(0, \frac{1}{\sqrt{5}}\right)$ (d) $\left(\frac{1}{\sqrt{5}}, \frac{1}{2}\right)$

Comprehension (51 and 52)

Let p, q_n be integers and let α, β be the roots of the equation, $x^2 - x - 1 = 0$ where $\alpha \neq \beta$. For $n = 0, 1, 2, \dots$ let $a_n = p\alpha^n + q\beta^n$.

FACT: If a and b are rational numbers and $a + b\sqrt{5} = 0$, then $a = b = 0$. (2017)

51. $a_{12} =$

(a) $a_{11} + 2a_{10}$ (b) $2a_{11} + a_{10}$
(c) $a_{11} - a_{10}$ (d) $a_{11} + a_{10}$

52. If $a_4 = 28$, then $p + 2q =$

(a) 14 (b) 7
(c) 21 (d) 12

53. Let α and β be the roots of $x^2 - x - 1 = 0$, with $\alpha > \beta$. For all positive integers n , define $a^n = \frac{\alpha^n - \beta^n}{\alpha - \beta}$, $n \geq 1$ (2019)

$b_1 = 1$ and $b_n = a_{n-1} + a_{n+1}$, $n \geq 2$.

Then which of the following options is/are correct?

(a) $a_1 + a_2 + a_3 + \dots + a_n = a_{n+2} - 1$ for all $n \geq 1$

(b) $\sum_{n=1}^{\infty} \frac{a_n}{10^n} = \frac{10}{89}$

(c) $\sum_{n=1}^{\infty} \frac{b_n}{10^n} = \frac{8}{89}$

(d) $b_n = \gamma^n + \beta^n$ for all $n \geq 1$

54. Suppose a, b denote the distinct real roots of the quadratic polynomial $x^2 + 20x - 2020$ and suppose c, d denote the distinct complex roots of the quadratic polynomial $x^2 - 20x + 2020$. Then the value of $ac(a-c) + ad(a-d) + bc(b-c) + bd(b-d)$ is (2021)

(a) 0 (b) 8000
(c) 8080 (d) 16000

ANSWER KEY

CONCEPT APPLICATION

1. (a) 2. (c) 3. (b) 4. (b) 5. (d) 6. (d) 7. (b) 8. (b) 9. (a) 10. (a)
 11. (a) 12. (a) 13. (i) $\left[\frac{3}{4}, \infty\right)$ (ii) $\left[\frac{31}{8}, \infty\right)$ (iii) $[-\infty, 8)$ (iv) $\left[-\frac{1}{4}, 8\right)$ (v) $\left[-1 - \frac{2}{\sqrt{3}}, -1 + \frac{2}{\sqrt{3}}\right]$
 (vi) $\left[1 - \frac{2}{\sqrt{3}}, 1 + \frac{2}{\sqrt{3}}\right]$ (vii) $(-\infty, 3 - \sqrt{8}] \cup [3 + \sqrt{8}, \infty)$ (viii) $(-\infty, -7 - \sqrt{48}] \cup [-7 + \sqrt{48}, \infty)$ (ix) $(-\infty, \infty)$ (x) $\left[-\frac{5}{3}, \frac{1}{2}\right]$
 (xi) $[1, \infty)$ (xii) $[4, 56]$ (xiii) $\left[\frac{1}{2}, 1\right]$ (xiv) $\left[\frac{1}{4}, 1\right]$ 14. (b) 15. (d) 16. (d) 17. $x = \pm\sqrt{7}$, $x = \pm 3$

EXERCISE-1 (TOPICWISE)

1. (c) 2. (d) 3. (a) 4. (c) 5. (a) 6. (d) 7. (a) 8. (c) 9. (b) 10. (a)
 11. (c) 12. (b) 13. (c) 14. (c) 15. (a) 16. (c) 17. (b) 18. (c) 19. (b) 20. (b)
 21. (c) 22. (a) 23. (a) 24. (c) 25. (d) 26. (b) 27. (c) 28. (c) 29. (b) 30. (d)
 31. (c) 32. (d) 33. (a) 34. (b) 35. (b) 36. (b) 37. (b) 38. (c) 39. (a) 40. (a)
 41. (c) 42. (b) 43. (b) 44. (b) 45. (b) 46. (d) 47. (b) 48. (b) 49. (a) 50. (b)
 51. (d) 52. (a) 53. (a) 54. (c) 55. (c) 56. (d) 57. (a) 58. (c) 59. (b) 60. (a)
 61. (d) 62. (b) 63. (d) 64. (c) 65. (c) 66. (b) 67. (c) 68. (c) 69. (a) 70. (a)
 71. (c) 72. (b) 73. (d) 74. (b) 75. (b) 76. (d) 77. (b) 78. (a) 79. (b) 80. (a)
 81. (b) 82. (a) 83. (a) 84. (c) 85. (a) 86. (d) 87. (b) 88. (a) 89. (b) 90. (b)

EXERCISE-2 (LEARNING PLUS)

1. (b) 2. (a) 3. (b) 4. (c) 5. (d) 6. (d) 7. (c) 8. (a) 9. (c) 10. (a)
 11. (b) 12. (a) 13. (c) 14. (a) 15. (d) 16. (d) 17. (a) 18. (c) 19. (c) 20. (c)
 21. (b) 22. (a) 23. (c) 24. (a) 25. (d) 26. (a) 27. (a) 28. (c) 29. (b) 30. (a)
 31. (c) 32. (a) 33. (a) 34. (a) 35. (d) 36. (a) 37. (a) 38. (b) 39. (d) 40. (b)
 41. (a) 42. (b) 43. (a) 44. (c) 45. (a) 46. (b) 47. (b) 48. (c) 49. (d) 50. (d)
 51. (a) 52. (d) 53. (d) 54. (b) 55. (d) 56. (c) 57. (d) 58. (d) 59. (c) 60. (c)
 61. $[1, 3]$ 62. $[13, 9]$ 63. $[6, -161/30]$ 64. $[7, 2]$ 65. $[(i) 1, 3, (ii) 1/2, 5 (iii) -3 (iv) a, a/2]$ 66. $[99]$
 67. $[*]$ 68. $[-2]$ 69. $[2]$ 70. $[-3/4, -3/5]$ 71. $[8]$ 72. $[7]$ 73. $[9]$ 74. $[23]$ 75. $[324]$
 76. $[191]$ 77. $[0]$ 78. $[2]$ 79. $(0, 3) \cup (3, 5)$

EXERCISE-3 (JEE ADVANCED LEVEL)

1. (a) 2. (b) 3. (b) 4. (c) 5. (c) 6. (c) 7. (a) 8. (a,d) 9. (b,c,d) 10. (a,b,c)
 11. (b,c,d) 12. (b,c) 13. (b,c,d) 14. (c) 15. (b) 16. (d) 17. (c) 18. (b) 19. (c) 20. (c)
 21. (a) 22. (c) 23. (c) 24. (b) 25. $[1]$ 26. $[10]$ 27. $[13]$ 28. $[7]$ 29. $[8]$ 30. $[2]$
 31. $[315]$ 32. $[3]$ 33. $[5]$ 34. $[3]$ 35. $[8]$ 36. $[16]$ 37. $[2]$ 38. $[*]$ 39. $[5]$ 40. $[4]$

EXERCISE-4 (PAST YEAR QUESTIONS)

JEE Main

1. (a) 2. (a) 3. (b) 4. (a) 5. (Bonus) 6. (c) 7. (d) 8. (d) 9. (c) 10. (b)
 11. (c) 12. (c) 13. (a) 14. (b) 15. (d) 16. (d) 17. (d) 18. (c) 19. (a) 20. (a)
 21. $[8]$ 22. (b) 23. (b) 24. (a) 25. (a) 26. (d) 27. (d) 28. (c) 29. (d) 30. (b)
 31. (b) 32. (d) 33. (b) 34. (d) 35. (a) 36. (d) 37. (c) 38. (a) 39. (b) 40. (c)
 41. (a) 42. $[0]$ 43. (b) 44. (b) 45. $[45]$ 46. $[98]$

JEE Advanced

47. (d) 48. (a,b,c) 49. (d) 50. (a, d) 51. (d) 52. (d) 53. (a,b,c) 54. (d)

